

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
2A	04	ENGINE 1 - PRE-HEATING RELAY FUEL FILTER	NO SIGNAL	Fuel filter pre-heater relay not working.		Check wiring and connections. Replace relay if required.				Battery goes flat.
2A	08	ENGINE 1 - PRE-HEATING RELAY FUEL FILTER	SIGNAL NOT PLAUSIBLE	Fuel filter pre-heater relay not working.		Check wiring and connections. Replace relay if required.				Battery goes flat.
2F	01	ENGINE 2 - GLOW PLUG RELAY	EXCEEDED UPPER LIMIT	Possible problematic cold cranking.	Short-circuit to positive, glow plugs always on also with ECU off, possible battery deployment.	Check wiring and connections. Replace relay if required.				
2F	02	ENGINE 2 - GLOW PLUG RELAY	BELOW LOWER LIMIT		Short-circuit to ground, glow plugs always on.	Check wiring and connections. Replace relay if required.				
2F	04	ENGINE 2 - GLOW PLUG RELAY	NO SIGNAL	Possible problematic cold cranking.	Faulty wiring.	Check wiring and connections. Replace relay if required.				Faulty diagnostic light.
2F	08	ENGINE 2 - GLOW PLUG RELAY	SIGNAL NOT PLAUSIBLE	Possible problematic cold cranking.	Faulty relay, wiring interrupted.	Check wiring and connections. Replace relay if required.				Possible increased fuel consumption. Possible engine overheating and power reduction.

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30	01	ENGINE 2 - COLD START LAMP	EXCEEDED UPPER LIMIT	Warning light always off. Problematic cold cranking. Pre-heater warning light always on.	Short-circuit to positive.	Check wiring and connections. Replace sensor if required.				The driver does not wait preheating even when the room temperatures are low, because no warning light signal is enabled. Preheating works, but with cold start-up no indication is available that tells you when to start the motor because the light is always turned on.
30	02	ENGINE 2 - COLD START LAMP	BELOW LOWER LIMIT	Warning light always off. Problematic cold cranking. Pre-heater warning light always on.	Short-circuit to ground.	Check wiring and connections. Replace sensor if required.				The driver does not wait preheating even when the room temperatures are low, because no warning light signal is enabled. Preheating works, but with cold start-up no indication is available that tells you when to start the motor because the light is always turned on.
30	04	ENGINE 2 - COLD START LAMP	NO SIGNAL	Warning light always off. Problematic cold cranking. Pre-heater warning light always on.		Check wiring and connections. Replace sensor if required.				Warning light off during pre-heating. Replace bulb if required.

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30	08	ENGINE 2 - COLD START LAMP	SIGNAL NOT PLAUSIBLE	Warning light always off. Problematic cold cranking. Pre-heater warning light always on.		Check wiring and connections. Replace sensor if required.				Warning light off during pre-heating. Replace bulb if required.
31	01	ENGINE 2 - CONTROL SYSTEM PRE-POSTHEATING	EXCEEDED UPPER LIMIT	Possible problematic cold cranking.	Short-circuit to positive.	Check wiring and connections. Check electrical system between relay and glow plugs.				Relay unit always on also with ECU off, possible battery deployment.
32	01	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	EXCEEDED UPPER LIMIT		Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
33	01	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	EXCEEDED UPPER LIMIT	The engine switching off-data are not memorized. The failures memory is lost, only the present failures and not the intermittent ones can be read, the idling speed, which can be eventually set by the Cruise Control commands, remains not memorized.	Faulty ECU EEPROM.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				

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33	02	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	BELOW LOWER LIMIT	The engine switching off-data are not memorized. The failures memory is lost, only the present failures and not the intermittent ones can be read, the idling speed, which can be eventually set by the Cruise Control commands, remains not memorized.	Faulty ECU EEPROM.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
33	04	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	NO SIGNAL	The engine switching off-data are not memorized. The failures memory is lost, only the present failures and not the intermittent ones can be read, the idling speed, which can be eventually set by the Cruise Control commands, remains not memorized.	Faulty ECU EEPROM.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				

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33	08	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	SIGNAL NOT PLAUSIBLE	The engine switching off-data are not memorized. The failures memory is lost, only the present failures and not the intermittent ones can be read, the idling speed, which can be eventually set by the Cruise Control commands, remains not memorized.	Faulty ECU EEPROM.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
34	08	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	SIGNAL NOT PLAUSIBLE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
35	08	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	SIGNAL NOT PLAUSIBLE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				

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36	08	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	SIGNAL NOT PLAUSIBLE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
37	01	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	EXCEEDED UPPER LIMIT		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
38	02	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	BELOW LOWER LIMIT		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
39	01	ENGINE 1 - AIR TEMPERATURE SENSOR	EXCEEDED UPPER LIMIT	Problematic cranking, smoke, problematic acceleration.		Check wiring and connections. Replace sensor if required.	Measure type: Resistance (KOhm) Measure point 1: Sensor Pin: 1 Measure point 2: Sensor Pin: 2	Connector Not connected; Key +15 OFF;	Typical Value: 2,5 KOhm;	Air temperature sensor and built-in pressure sensor. The sensor is fitted on flow meter in engines with EGR.

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39	02	ENGINE 1 - AIR TEMPERATURE SENSOR	BELOW LOWER LIMIT	Problematic cranking, smoke, problematic acceleration.	Short-circuit to ground, excessively high temperature is detected.	Check wiring and connections. Replace sensor if required.	Measure type: Resistance (KOhm) Measure point 1: Sensor Pin: 1 Measure point 2: Sensor Pin: 2	Connector Not connected; Key +15 OFF;	Typical Value: 2,5 KOhm;	Air temperature sensor and built-in pressure sensor. The sensor is fitted on flow meter in engines with EGR.
3A	02	ELECTRONIC CONTROL UNIT - IMMOBILISER	BELOW LOWER LIMIT	The engine fails to start	Communication with Immobilizer ECU problems on CAN Line.	Check integrity of CAN Line, run Immobilizer ECU diagnostics and wait for indications provided.	Measure type: Resistance (Ohm) Measure point 1: Diagnostic socket. Pin: 21 Measure point 2: Diagnostic socket. Pin: 22	Connector Connected; Key +15 OFF;	Typical Value: 60 Ohm Ohm;	
3C	01	INJECTOR BENCH 1	- EXCEEDED UPPER LIMIT	Engine not working properly, possible power reduction.	Injector wiring short-circuit.	Check wiring and connections. Replace injector if required.				Only two cylinders running.
3C	02	INJECTOR BENCH 1	- BELOW LOWER LIMIT	Engine not working properly, possible power reduction.	Short-circuit to ground.	Check wiring and connections.				Only two cylinders running.
3C	08	INJECTOR BENCH 1	- SIGNAL NOT PLAUSIBLE	Engine not working properly, possible power reduction.	Injector electrical system failure.	Check wiring and connections. Replace injector if required.				Only two cylinders running.
3D	04	INJECTOR BENCH 1	- NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring disconnected.	Check wiring and connections. Replace injector if required.				Only two cylinders running.
3E	01	INJECTOR BENCH 2	- EXCEEDED UPPER LIMIT	Engine not working properly, possible power reduction.	Injector wiring short-circuit.	Check wiring and connections. Replace injector if required.				Only two cylinders running.

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
3E	02	INJECTOR BENCH 2	-	BELOW LOWER LIMIT	Engine not working properly, possible power reduction.	Short-circuit to ground.	Check wiring and connections.			Only two cylinders running.
3E	08	INJECTOR BENCH 2	-	SIGNAL NOT PLAUSIBLE	Engine not working properly, possible power reduction.	Injector electrical system failure.	Check wiring and connections. Replace injector if required.			Only two cylinders running.
3F	04	INJECTOR BENCH 2	-	NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring disconnected.	Check wiring and connections. Replace injector if required.			Only two cylinders running.
40	01	INJECTOR INJECTOR SUPPLY	- A	EXCEEDED UPPER LIMIT	Engine off.	Internal ECU problem.	Clear failure memory. If the error persists, ensure that the failure does not concern the injectors and call the Help Desk for instructions on how to replace the ECU.			
40	02	INJECTOR INJECTOR SUPPLY	- A	BELOW LOWER LIMIT	Engine off.	Internal ECU problem.	Clear failure memory. If the error persists, ensure that the failure does not concern the injectors and call the Help Desk for instructions on how to replace the ECU.			

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40	04	INJECTOR INJECTOR SUPPLY	- A	NO SIGNAL	Engine off.	Internal ECU problem.	Clear failure memory. If the error persists, ensure that the failure does not concern the injectors and call the Help Desk for instructions on how to replace the ECU.				
40	08	INJECTOR INJECTOR SUPPLY	- A	SIGNAL NOT PLAUSIBLE	Engine off.	Internal ECU problem.	Clear failure memory. If the error persists, ensure that the failure does not concern the injectors and call the Help Desk for instructions on how to replace the ECU.				
41	01	INJECTOR INJECTOR SUPPLY	- B	EXCEEDED UPPER LIMIT	Engine off.	Internal ECU problem.	Clear failure memory. If the error persists, ensure that the failure does not concern the injectors and call the Help Desk for instructions on how to replace the ECU.				

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
41	02	INJECTOR INJECTOR SUPPLY	- B BELOW LOWER LIMIT	Engine off.	Internal ECU problem.	Clear failure memory. If the error persists, ensure that the failure does not concern the injectors and call the Help Desk for instructions on how to replace the ECU.				
41	04	INJECTOR INJECTOR SUPPLY	- B NO SIGNAL	Engine off.	Internal ECU problem.	Clear failure memory. If the error persists, ensure that the failure does not concern the injectors and call the Help Desk for instructions on how to replace the ECU.				
41	08	INJECTOR INJECTOR SUPPLY	- B SIGNAL NOT PLAUSIBLE	Engine off.	Internal ECU problem.	Clear failure memory. If the error persists, ensure that the failure does not concern the injectors and call the Help Desk for instructions on how to replace the ECU.				
42	01	INJECTOR INJECTOR I	- EXCEEDED UPPER LIMIT	Engine not working properly, possible power reduction.	Short-circuit to positive.	Check wiring and connections. Replace injector if required.				Only three cylinders running.

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42	01	INJECTOR INJECTOR I	- EXCEEDED UPPER LIMIT				1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A47 Measure point 2: Injector Pin: 2 2- Measure type: Resistance (Ohm) Measure point 1: Injector Pin: 1 Measure point 2: Injector Pin: 2 3- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A16 Measure point 2: Injector Pin: 1	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0.1 Ohm; 2- Min. value: 0.5 Ohm; Max. value: 0.9 Ohm; Typical Value: 0.7 Ohm; 3- Typical Value: 0,1 Ohm;	
42	04	INJECTOR INJECTOR I	- NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring short-circuit.	Check wiring and connections. Replace injector if required.				Only two cylinders running.
42	04	INJECTOR INJECTOR I	- NO SIGNAL				1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A47 Measure point 2: Injector Pin: 2 2- Measure type: Resistance (Ohm) Measure point 1: Injector Pin: 1 Measure point 2: Injector Pin: 2 3- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A16 Measure point 2: Injector Pin: 1	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0.1 Ohm; 2- Min. value: 0.5 Ohm; Max. value: 0.9 Ohm; Typical Value: 0.7 Ohm; 3- Typical Value: 0,1 Ohm;	

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42	08	INJECTOR INJECTOR 1	- SIGNAL NOT PLAUSIBLE	Engine not working properly, possible power reduction.	Injector not working properly.	Check wiring and connections. Replace injector if required.				Only three cylinders running.
42	08	INJECTOR INJECTOR 1	- SIGNAL NOT PLAUSIBLE				1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A47 Measure point 2: Injector Pin: 2 2- Measure type: Resistance (Ohm) Measure point 1: Injector Pin: 1 Measure point 2: Injector Pin: 2 3- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A16 Measure point 2: Injector Pin: 1	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0.1 Ohm; 2- Min. value: 0.5 Ohm; Max. value: 0.9 Ohm; Typical Value: 0.7 Ohm; 3- Typical Value: 0,1 Ohm;	
43	04	INJECTOR INJECTOR 1	- NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring open circuit.	Check wiring and connections. Replace injector if required.				Only three cylinders running.
44	01	INJECTOR INJECTOR 2	- EXCEEDED UPPER LIMIT	Engine not working properly, possible power reduction.	Short-circuit to positive.	Check wiring and connections. Replace injector if required.				Only three cylinders running.

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44	01	INJECTOR INJECTOR 2	- EXCEEDED UPPER LIMIT				1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A17 Measure point 2: Injector Pin: 1 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A13 Measure point 2: Injector Pin: 2 3- Measure type: Resistance (Ohm) Measure point 1: Injector Pin: 1 Measure point 2: Injector Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	
44	04	INJECTOR INJECTOR 2	- NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring short-circuit.	Check wiring and connections. Replace injector if required.				Only two cylinders running.
44	04	INJECTOR INJECTOR 2	- NO SIGNAL				1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A17 Measure point 2: Injector Pin: 1 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A13 Measure point 2: Injector Pin: 2 3- Measure type: Resistance (Ohm) Measure point 1: Injector Pin: 1 Measure point 2: Injector Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	

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44	08	INJECTOR INJECTOR 2	- SIGNAL NOT PLAUSIBLE	Engine not working properly, possible power reduction.	Injector not working properly.	Check wiring and connections. Replace injector if required.				Only three cylinders running.
44	08	INJECTOR INJECTOR 2	- SIGNAL NOT PLAUSIBLE				1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A17 Measure point 2: Injector Pin: 1 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A13 Measure point 2: Injector Pin: 2 3- Measure type: Resistance (Ohm) Measure point 1: Injector Pin: 1 Measure point 2: Injector Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	
45	04	INJECTOR INJECTOR 2	- NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring open circuit.	Check wiring and connections. Replace injector if required.				Only three cylinders running.
46	01	INJECTOR INJECTOR 3	- EXCEEDED UPPER LIMIT	Engine not working properly, possible power reduction.	Short-circuit to positive.	Check wiring and connections. Replace injector if required.				Only three cylinders running.
46	04	INJECTOR INJECTOR 3	- NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring short-circuit.	Check wiring and connections. Replace injector if required.				Only two cylinders running.

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46	08	INJECTOR INJECTOR 3	- SIGNAL NOT PLAUSIBLE	Engine not working properly, possible power reduction.	Injector not working properly.	Check wiring and connections. Replace injector if required.				Only three cylinders running.
47	04	INJECTOR INJECTOR 3	- NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring open circuit.	Check wiring and connections. Replace injector if required.				Only three cylinders running.
47	04	INJECTOR INJECTOR 3	- NO SIGNAL				1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A3I Measure point 2: Injector Pin: 2 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: AI Measure point 2: Injector Pin: I 3- Measure type: Resistance (Ohm) Measure point 1: Injector Pin: I Measure point 2: Injector Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	
48	01	INJECTOR INJECTOR 4	- EXCEEDED UPPER LIMIT	Engine not working properly, possible power reduction.	Short-circuit to positive.	Check wiring and connections. Replace injector if required.				Only three cylinders running.
48	04	INJECTOR INJECTOR 4	- NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring short-circuit.	Check wiring and connections. Replace injector if required.				Only two cylinders running.

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48	08	INJECTOR INJECTOR 4	- SIGNAL NOT PLAUSIBLE	Engine not working properly, possible power reduction.	Injector not working properly.	Check wiring and connections. Replace injector if required.				Only three cylinders running.
49	04	INJECTOR INJECTOR 4	- NO SIGNAL	Engine not working properly, possible power reduction.	Injector wiring open circuit.	Check wiring and connections. Replace injector if required.				Only three cylinders running.
49	04	INJECTOR INJECTOR 4	- NO SIGNAL				1- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A46 Measure point 2: Injector Pin: 2 2- Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A1 Measure point 2: Injector Pin: 1 3- Measure type: Resistance (Ohm) Measure point 1: Injector Pin: 1 Measure point 2: Injector Pin: 2	1- Connector Not connected; Key +15 OFF; 2- Connector Not connected; Key +15 OFF; 3- Connector Not connected; Key +15 OFF;	1- Typical Value: 0,1 Ohm; 2- Typical Value: 0,1 Ohm; 3- Min. value: 0,5 Ohm; Max. value: 0,9 Ohm; Typical Value: 0,7 Ohm;	
4E	08	VEHICLE CRUISE CONTROL SWITCH UNIT	- SIGNAL NOT PLAUSIBLE	Cruise control / PTO not working.	Press SET+ / SET- and RESUME/ OFF at the same time.	Check correct operation of the switch by reading state parameters.				Replace wiring and connections if state does not change when Cruise Control buttons are pressed.

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50	01	ELECTRONIC CONTROL UNIT - MAIN RELAY DEFECT	EXCEEDED UPPER LIMIT	Engine does not start, ECU not powered or ECU always powered and EDC off also at key-on.	Main relay interrupted or short-circuit.	Check wiring and connections. Replace relay if required.				
50	02	ELECTRONIC CONTROL UNIT - MAIN RELAY DEFECT	BELOW LOWER LIMIT	Engine does not start, ECU not powered or ECU always powered and EDC off also at key-on.	Main relay interrupted or short-circuit.	Check wiring and connections. Replace relay if required.				
51	01	VEHICLE - MULTIPOSITION SELECTOR / PTO	EXCEEDED UPPER LIMIT	Incorrect PTO operation.	Voltage exceeding max. threshold, short-circuit to positive.	Check wiring and connections. Replace sensor if required.				
51	02	VEHICLE - MULTIPOSITION SELECTOR / PTO	BELOW LOWER LIMIT	Incorrect PTO operation.	Voltage under min. threshold, short-circuit to ground.	Check wiring and connections. Replace sensor if required.				
51	08	VEHICLE - MULTIPOSITION SELECTOR / PTO	SIGNAL NOT PLAUSIBLE	Incorrect PTO operation.	Faulty device.	Check wiring and connections. Replace sensor if required.				

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52	04	FUEL PRESSURE - PRESSURE MPROP REGULATOR ERROR	NO SIGNAL	Engine off.	Faulty MPROP.	Check wiring and connections.	Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A49 Measure point 2: ECU Pin: A19	Connector Not connected; Key +15 OFF;	Min. value: 3.2 Ohm; Max. value: 3.6 Ohm; Typical Value: 3.4 Ohm;	High noise.
52	08	FUEL PRESSURE - PRESSURE MPROP REGULATOR ERROR	SIGNAL NOT PLAUSIBLE			Check wiring and connections. Replace ECU if required.	Measure type: Resistance (Ohm) Measure point 1: ECU Pin: A49 Measure point 2: ECU Pin: A19	Connector Not connected; Key +15 OFF;	Min. value: 3.2 Ohm; Max. value: 3.6 Ohm; Typical Value: 3.4 Ohm;	
53	01	FUEL PRESSURE - PRESSURE MPROP REGULATOR ERROR (SHORT CIRCUIT TO POSITIVE)	EXCEEDED UPPER LIMIT		Short-circuit to battery, faulty MPROP.	Check wiring and connections. Replace MPROP if required.				
54	01	FUEL PRESSURE - PRESSURE MPROP REGULATOR ERROR (SHORT CIRCUIT TO NEGATIVE)	EXCEEDED UPPER LIMIT		Short-circuit to ground. Faulty MPROP.	Check wiring and connections. Replace MPROP if required.				
56	08	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	SIGNAL NOT PLAUSIBLE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				

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5A	01	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	EXCEEDED UPPER LIMIT		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
5A	02	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	BELOW LOWER LIMIT		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
5B	01	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	EXCEEDED UPPER LIMIT		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				

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5E	01	ENGINE I - FUEL PUMP RELAY	EXCEEDED UPPER LIMIT	Fuel pump on always when engine is off.	Faulty relay, short-circuit to positive in wiring.	Turn key-on: pump must run for approximately 10 seconds (it should hum). Check pump relay if pump remains on. Check wiring if all checks are OK.				
5E	02	ENGINE I - FUEL PUMP RELAY	BELOW LOWER LIMIT	Fuel pump not working.	Faulty relay, short-circuit to ground in wiring.	Turn key-on: pump must run for approximately 10 seconds (it should hum). Check the pump relay, protection fuse and wiring if this does not occur.				
5E	04	ENGINE I - FUEL PUMP RELAY	NO SIGNAL	Fuel pump not working.	Faulty relay, wiring interrupted.	Check wiring and connections. Replace relay if required.				
5E	08	ENGINE I - FUEL PUMP RELAY	SIGNAL NOT PLAUSIBLE	Fuel pump not working.	Faulty relay, wiring interrupted.	Check wiring and connections. Replace relay if required.				
5F	01	FUEL PRESSURE - RAIL PRESSURE SENSOR OR SIGNAL ERROR	EXCEEDED UPPER LIMIT		Short-circuit to positive. Faulty sensor. Rail pressure not regular.	Check wiring and connections. Replace sensor if required.				Check DTC I 03 error.

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
5F	02	FUEL PRESSURE - RAIL PRESSURE SENSOR OR SIGNAL ERROR	BELOW LOWER LIMIT		Short-circuit to ground, faulty sensor.	Check wiring and connections. Replace sensor if required.				
60	01	FUEL PRESSURE - RAIL PRESSURE SENSOR OFFSET	EXCEEDED UPPER LIMIT		Faulty rail pressure sensor.	Replace sensor.				
60	02	FUEL PRESSURE - RAIL PRESSURE SENSOR OFFSET	BELOW LOWER LIMIT		Faulty rail pressure sensor.	Replace sensor.				
62	01	FUEL PRESSURE - FAULT ON THE FUEL DRUCK CONTROL OF THE RAIL (POSITIVE DEVIATION)	EXCEEDED UPPER LIMIT		High pressure circuit fuel leakage.	Check fuel feed system.				Fuel management and pressure failure in rail.
62	01	FUEL PRESSURE - FAULT ON THE FUEL DRUCK CONTROL OF THE RAIL (POSITIVE DEVIATION)	EXCEEDED UPPER LIMIT		Injector jammed in fuel passage open position.	Check hydraulic and mechanical efficiency of injectors.				Fuel management and pressure failure in rail.

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
62	01	FUEL PRESSURE - FAULT ON THE FUEL DRUCK CONTROL OF THE RAIL (POSITIVE DEVIATION)	EXCEEDED UPPER LIMIT		MPROP adjuster open movement jammed.	Check efficiency of MPROP adjuster.				Fuel management and pressure failure in rail.
62	01	FUEL PRESSURE - FAULT ON THE FUEL DRUCK CONTROL OF THE RAIL (POSITIVE DEVIATION)	EXCEEDED UPPER LIMIT		Faulty high pressure pump.	Check efficiency of high pressure pump.				Fuel management and pressure failure in rail.
63	01	FUEL PRESSURE - FAULT ON THE FUEL DRUCK CONTROL OF THE RAIL (NEGATIVE DEVIATION)	EXCEEDED UPPER LIMIT		MPROP adjuster open movement jammed.	Check efficiency of MPROP adjuster.				Fuel management and pressure failure in rail.
64	01	FUEL PRESSURE - RAIL PRESSURE ERROR: TOO LOW	EXCEEDED UPPER LIMIT		High pressure circuit fuel leakage.	Check high pressure system. Replace high pressure pump if required.				Fuel management and pressure failure in rail.
65	01	FUEL PRESSURE - RAIL PRESSURE ERROR: TOO HIGH	EXCEEDED UPPER LIMIT		MPROP regulator jammed.	Check MPROP regulator, replace if required.				

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
66	01	FUEL PRESSURE - ERROR ON THE RAIL PRESSURE (EXCESSIVE DUTY CYCLE)	EXCEEDED UPPER LIMIT	Negative vehicle reaction with smoke in exhaust during acceleration.	High pressure circuit fuel leakage.	Check fuel feed system, replace high pressure pump if required. Faulty fuel feed system (fuel pump and filter jammed).				
67	01	FUEL PRESSURE - ERROR ON THE RAIL PRESSURE (EXCESSIVE)	EXCEEDED UPPER LIMIT	Engine off.	MPROP regulator jammed.	Check MPROP regulator, replace if required.				Replace pressure relief valve.
68	02	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	BELOW LOWER LIMIT			Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
68	04	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	NO SIGNAL		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
68	08	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	SIGNAL NOT PLAUSIBLE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
69	01	ELECTRONIC CONTROL UNIT - SENSOR POWER SUPPLY	EXCEEDED UPPER LIMIT	Anomalous engine operation due to incorrectly powered sensors. Reduced power.	Sensor power circuit fault in ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				Possible fault indications of various sensors powered by ECU.
69	02	ELECTRONIC CONTROL UNIT - SENSOR POWER SUPPLY	BELOW LOWER LIMIT	Anomalous engine operation due to incorrectly powered sensors. Reduced power.	Sensor power circuit fault in ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				Possible fault indications of various sensors powered by ECU.

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
6A	01	ELECTRONIC CONTROL UNIT - SENSOR POWER SUPPLY	EXCEEDED UPPER LIMIT	Anomalous engine operation due to incorrectly powered sensors. Reduced power.	Sensor power circuit fault in ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				Possible fault indications of various sensors powered by ECU.
6A	02	ELECTRONIC CONTROL UNIT - SENSOR POWER SUPPLY	BELOW LOWER LIMIT	Anomalous engine operation due to incorrectly powered sensors. Reduced power.	Sensor power circuit fault in ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				Possible fault indications of various sensors powered by ECU.
6B	01	ELECTRONIC CONTROL UNIT - SENSOR POWER SUPPLY	EXCEEDED UPPER LIMIT	Anomalous engine operation due to incorrectly powered sensors. Reduced power.	Sensor power circuit fault in ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				Possible fault indications of various sensors powered by ECU.

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
6B	02	ELECTRONIC CONTROL UNIT - SENSOR POWER SUPPLY	BELOW LOWER LIMIT	Anomalous engine operation due to incorrectly powered sensors. Reduced power.	Sensor power circuit fault in ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				Possible fault indications of various sensors powered by ECU.
6C	01	VEHICLE - EDC LAMP	EXCEEDED UPPER LIMIT	Warning light not working.	Short-circuit to positive.	Check correct operation of warning light using "Active diagnostic" procedure.				Warning light should come on for approximately 5 seconds at key-on. Check wiring and connections if this does not occur.
6C	02	VEHICLE - EDC LAMP	BELOW LOWER LIMIT	Warning light not working.	Short-circuit to ground.	Check correct operation of warning light using "Active diagnostic" procedure.				Warning light should come on for approximately 5 seconds at key-on. Check wiring and connections if this does not occur.
6C	04	VEHICLE - EDC LAMP	NO SIGNAL	Warning light not working.	Open circuit, bulb disconnected.	Check correct operation of warning light using "Active diagnostic" procedure.				Warning light should come on for approximately 5 seconds at key-on. Check wiring and connections if this does not occur.

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
6C	08	VEHICLE - EDC LAMP	SIGNAL NOT PLAUSIBLE	Warning light not working.	Wiring problems.	Check wiring and connections. Replace sensor if required.				Warning light should come on for approximately 5 seconds at key-on. Check wiring and connections if this does not occur.
6D	08	ENGINE 2 - INTERNAL ECU FAULT (PLAUSIBILITY ERROR +15)	SIGNAL NOT PLAUSIBLE			Check wiring and connections.				Key 15 off during initialisation.
6E	08	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	SIGNAL NOT PLAUSIBLE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				
75	01	VEHICLE - VEHICLE SPEED SENSOR / SIGNAL	EXCEEDED UPPER LIMIT	Speed of 170 km/h exceeded.		Check correct calibration of speedometer.				Encourage driver to use the vehicle correctly.
75	04	VEHICLE - VEHICLE SPEED SENSOR / SIGNAL	NO SIGNAL		Interrupted wiring between vehicle speed sensor and instrument panel.	Check wiring and connections between vehicle speed sensor and instrument panel.				
75	04	VEHICLE - VEHICLE SPEED SENSOR / SIGNAL	NO SIGNAL		Wiring interrupted between instrument panel and EDC ECU.	Check wiring and connections between instrument panel and EDC ECU.				Intervention required if instrument panel indicates vehicle speed.

DTC	FMI	Failing component	Type of Failure	Visible failure	Possible Cause	Repair action	Checks to be performed	Measuring conditions	Values to be detected	Remarks
75	04	VEHICLE SPEED SENSOR / - SIGNAL	NO SIGNAL		Vehicle speed sensor disconnected or failed.	Check correct assembly and efficiency of vehicle speed sensor.				
75	08	VEHICLE SPEED SENSOR / - SIGNAL	SIGNAL NOT PLAUSIBLE		Vehicle speed sensor disconnected or failed.	Check correct assembly and efficiency of vehicle speed sensor.				
75	08	VEHICLE SPEED SENSOR / - SIGNAL	SIGNAL NOT PLAUSIBLE	Vehicle speed on instrument panel does not increase sensibly.	Wrong speedometer setting.	Check correct calibration of speedometer.				
77	01	VEHICLE SPEED SENSOR / - SIGNAL	EXCEEDED UPPER LIMIT	Wrong vehicle speed indication.	Wrong speedometer setting.	Check correct calibration of speedometer.				
77	02	VEHICLE SPEED SENSOR / - SIGNAL	BELOW LOWER LIMIT	Wrong vehicle speed indication.	Wrong speedometer setting.	Check correct calibration of speedometer.				
77	08	VEHICLE SPEED SENSOR / - SIGNAL	SIGNAL NOT PLAUSIBLE	Wrong vehicle speed indication.	Wrong speedometer setting.	Check correct calibration of speedometer.				
79	08	ELECTRONIC CONTROL UNIT - INTERNAL ECU FAULT	SIGNAL NOT PLAUSIBLE		Wrong ECU programming. Probable electromagnetic interference. Faulty ECU.	Switch key on/off and wait for a few seconds, clear failure memory. If the error persists, call the Help Desk for instructions on how to replace the ECU.				

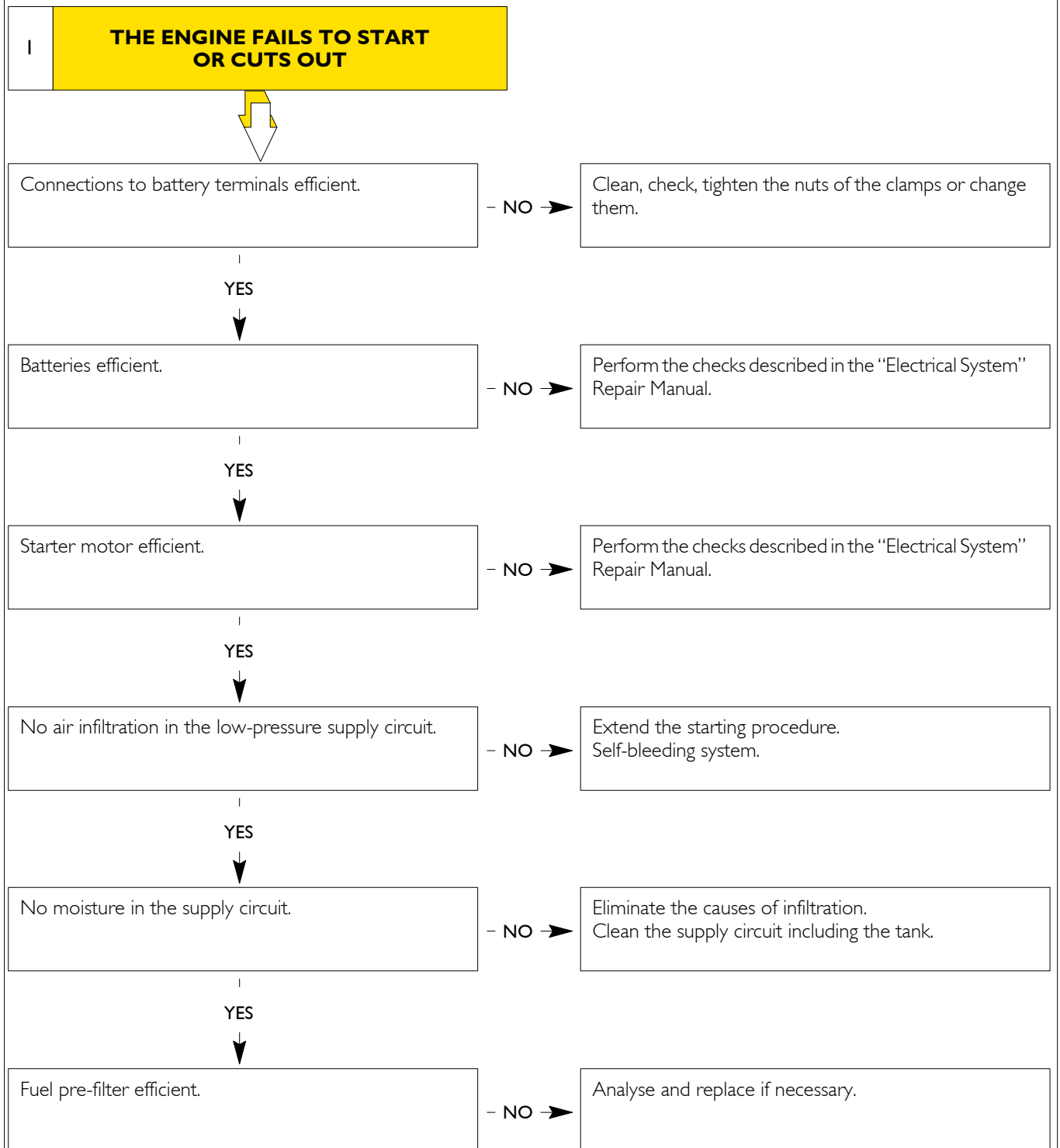
2nd Section

SYMPTOMS

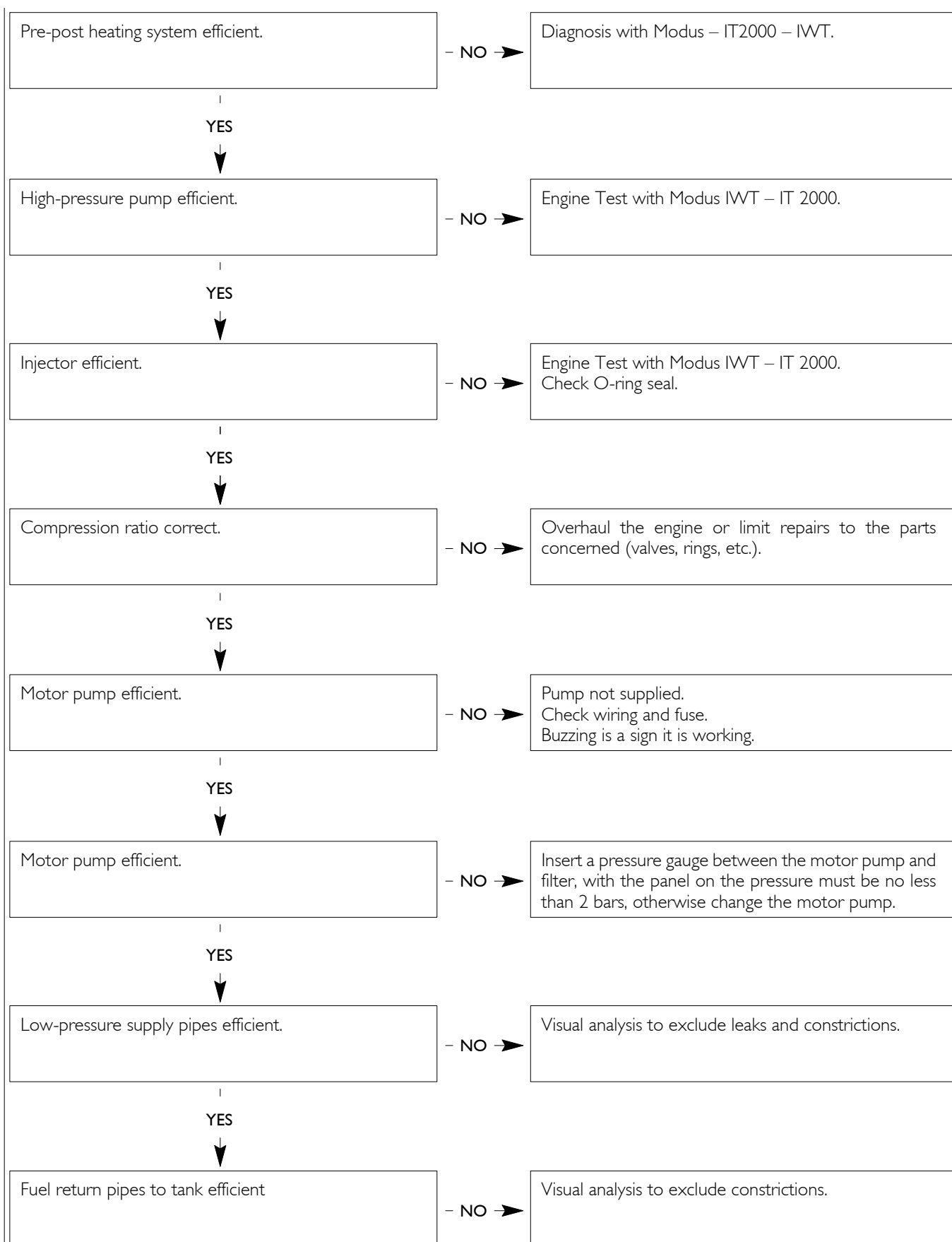
Main engine operating trouble:

- 1 – The engine fails to start or cuts out;
- 2 – The engine overheats;
- 3 – The engine lacks power;
- 4 – The engine produces black or dark grey smoke;
- 5 – The engine produces grey (tending to white) smoke;

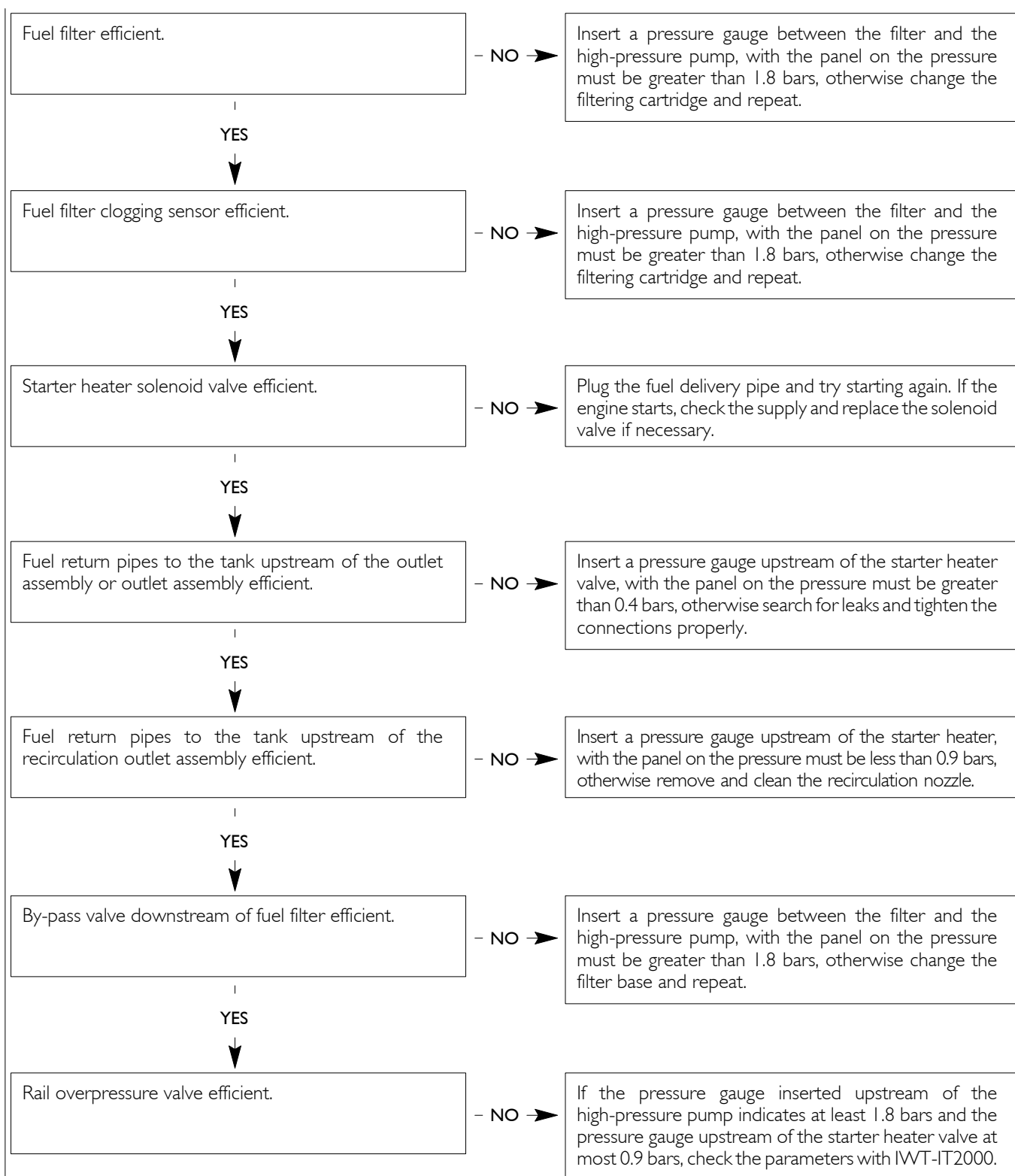
- 6 – The engine produces blue smoke;
- 7 – The engine produces white or black smoke;
- 8 – The engine knocks abnormally;
- 9 – The engine stops;
- 10 – Fuel consumption too high;
- 11 – Oil pressure low or too high.



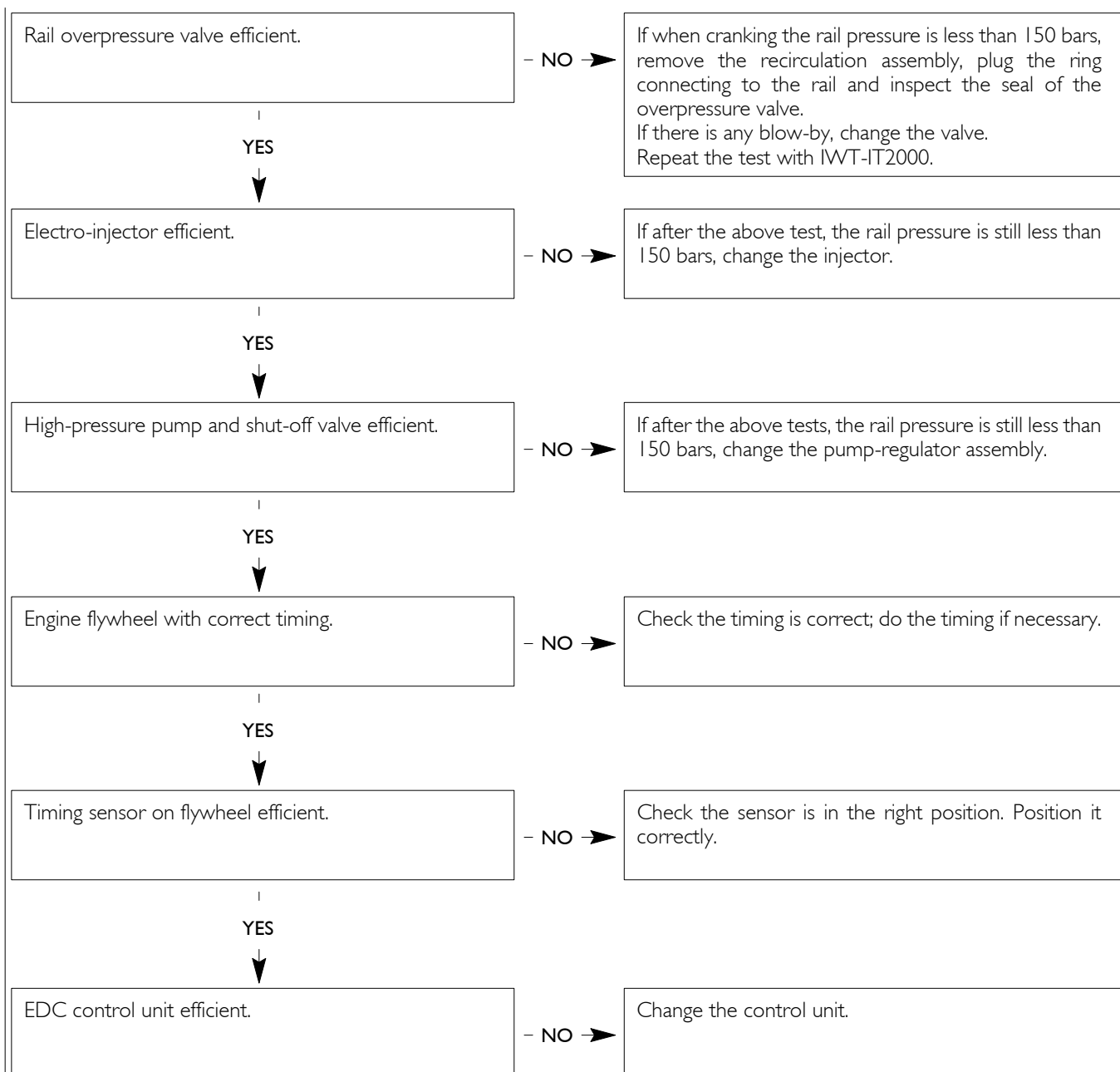
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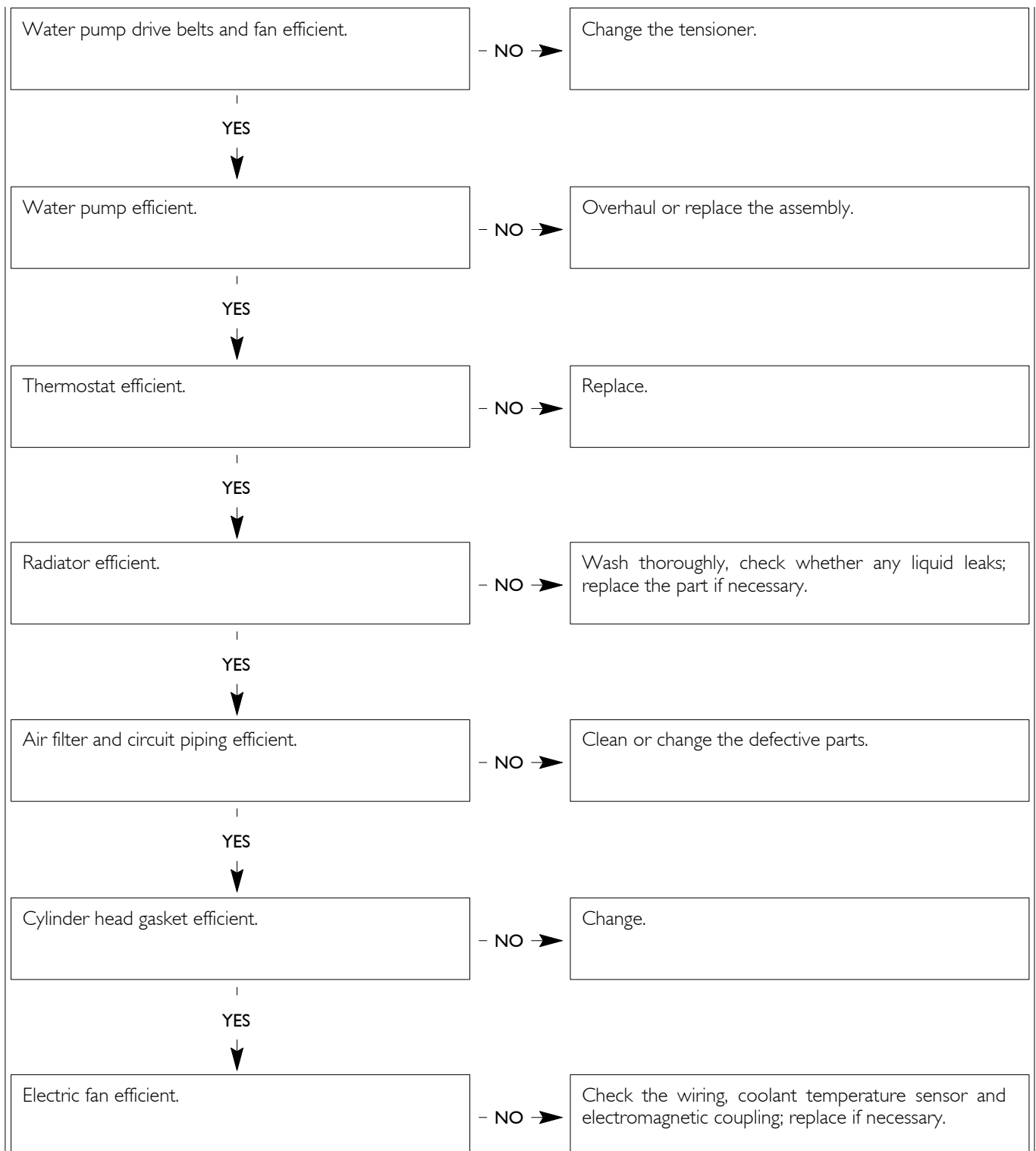


Correct coolant level.

- NO →

Check for leaks and restore the level.

(continues)



3

THE ENGINE LACKS POWER

Fuel filters efficient.

- NO →

Change, proceeding as described in the "Operation and Maintenance" booklet.

YES



High and low pressure, supply circuit efficient.

- NO →

Check system coupling seal.

YES



Thermostat efficient.

- NO →

Change.

YES



Fuel in tank.

- NO →

Check the fuel level.

YES



Drawing element of fuel in tank efficient.

- NO →

Check the priming pump and clean the drawing element of any debris.

YES



High-pressure pump efficient.

- NO →

Engine Test with Modus – IWT – IT2000, replace if necessary.

YES

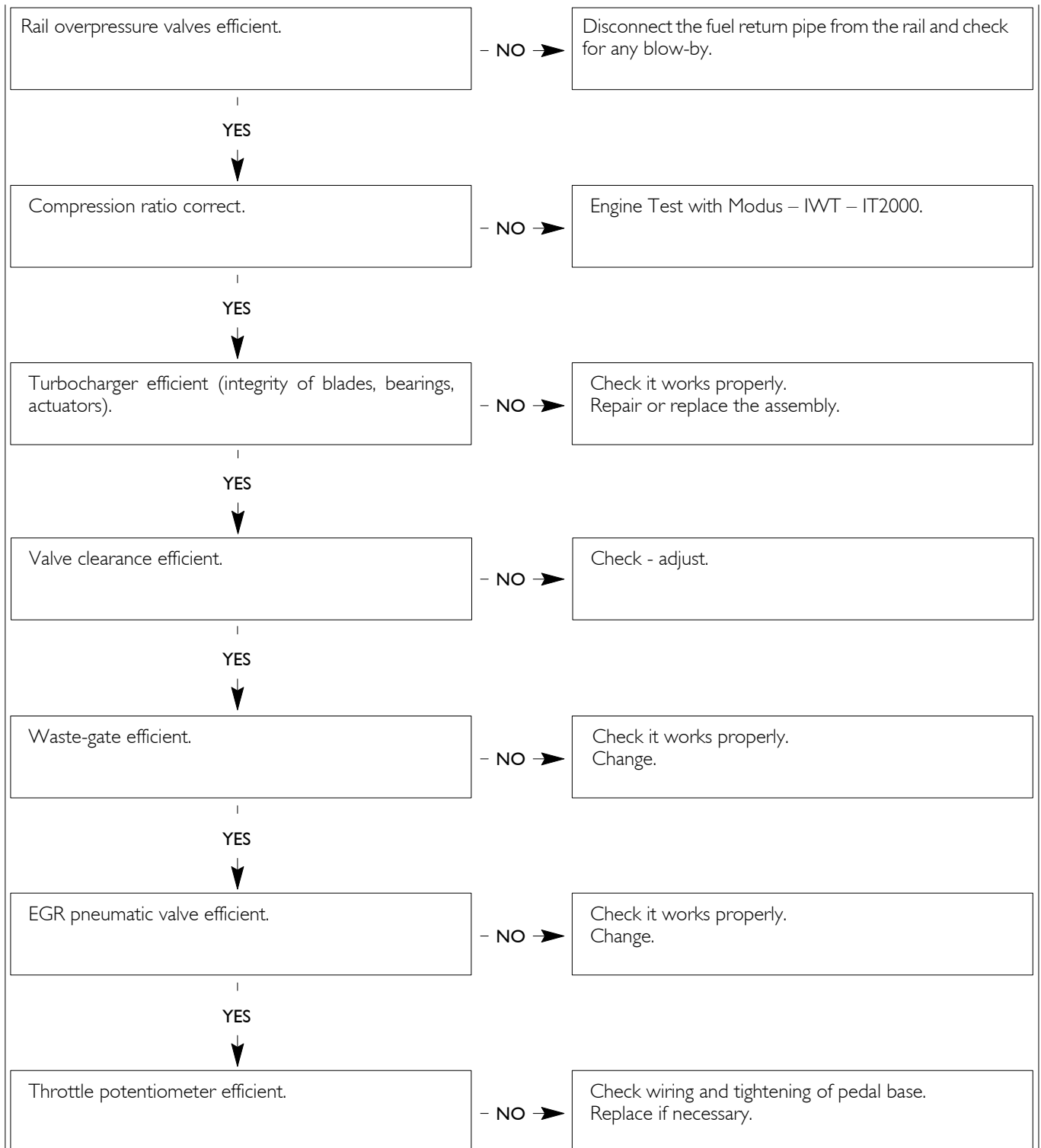


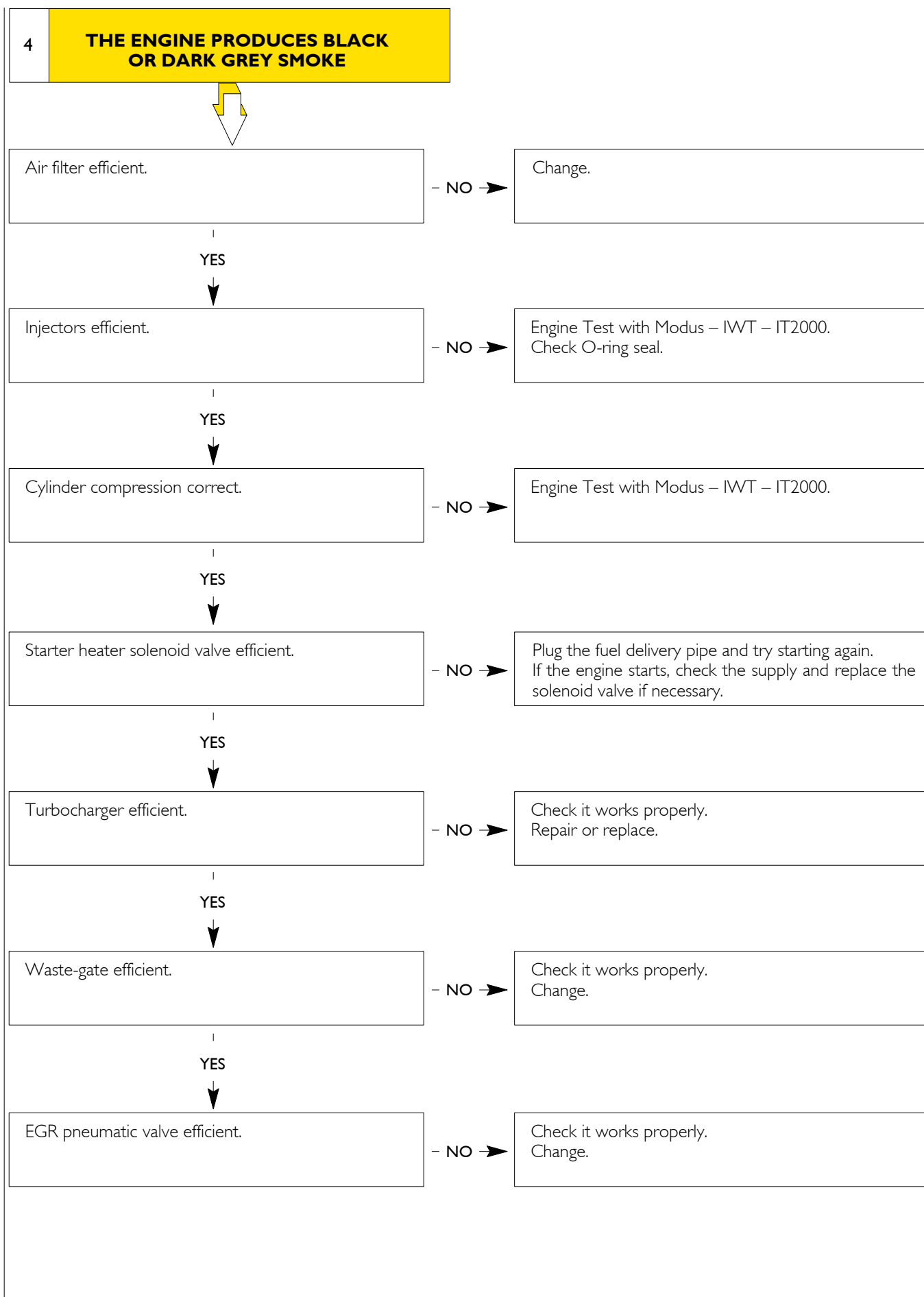
Injectors efficient and flow limiters (if applicable) work correctly

- NO →

Engine Test with Modus – IWT – IT2000, replace if necessary.
Check O-ring seal.

(continues)





5

THE ENGINE PRODUCES GREY SMOKE, TENDING TO WHITE

Injectors efficient.

- NO →

Engine Test with Modus – IWT IT 2000.
Check O-ring seal.

YES



Coolant level correct.

- NO →

Probable passage of coolant into the combustion chambers. Change the cylinder head gasket or overhaul the engine.

6

THE ENGINE PRODUCES BLUE SMOKE

Oil in the cylinders.

- YES →

Too much oil consumption.
Turbine oil blow-by.
Blow-by from valve guide.

7

THE ENGINE PRODUCES WHITE OR BLACK SMOKE

Timing sensor on flywheel efficient.

- YES →

Check the sensor is correctly positioned.
Position it correctly.

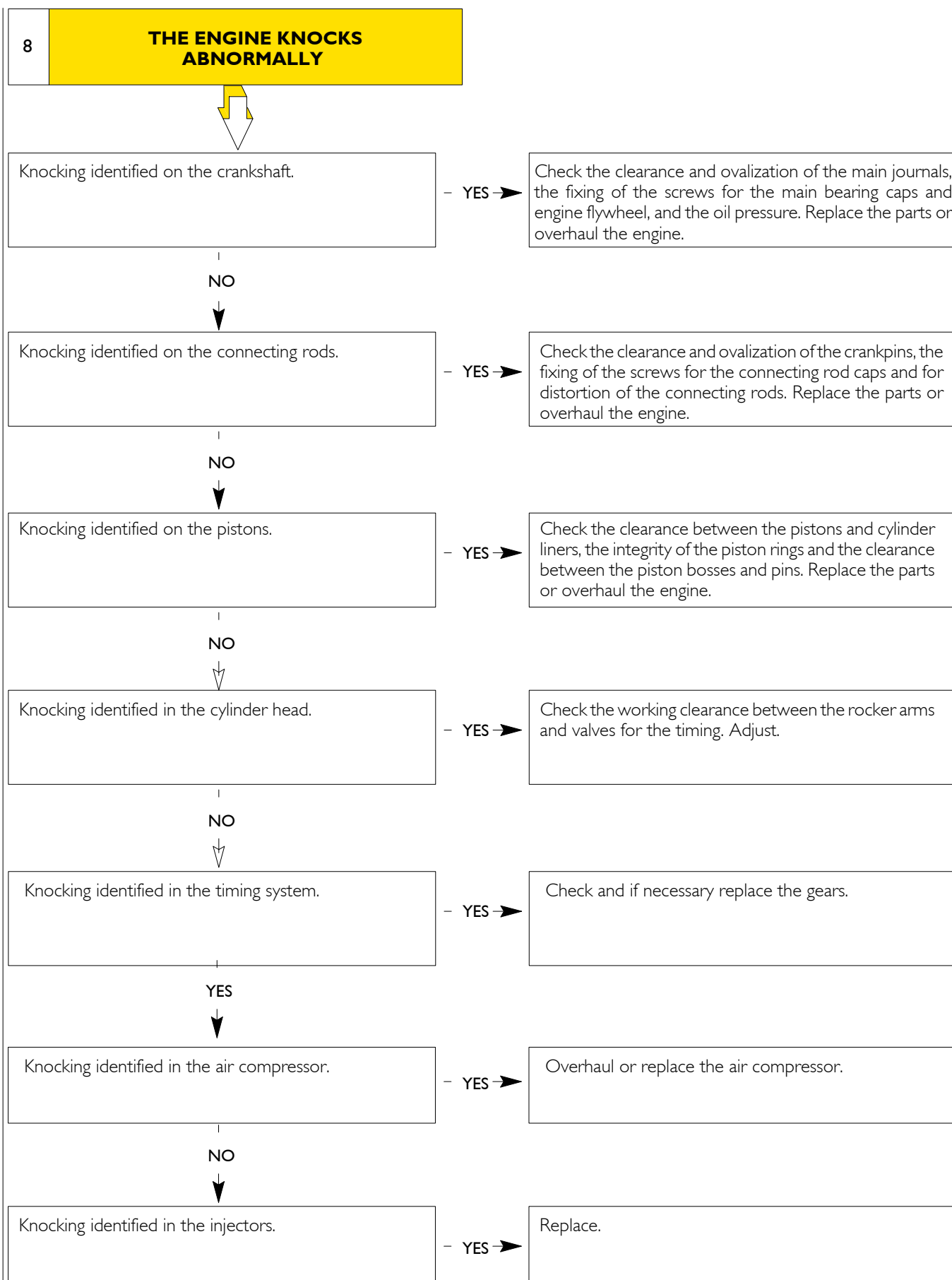
YES



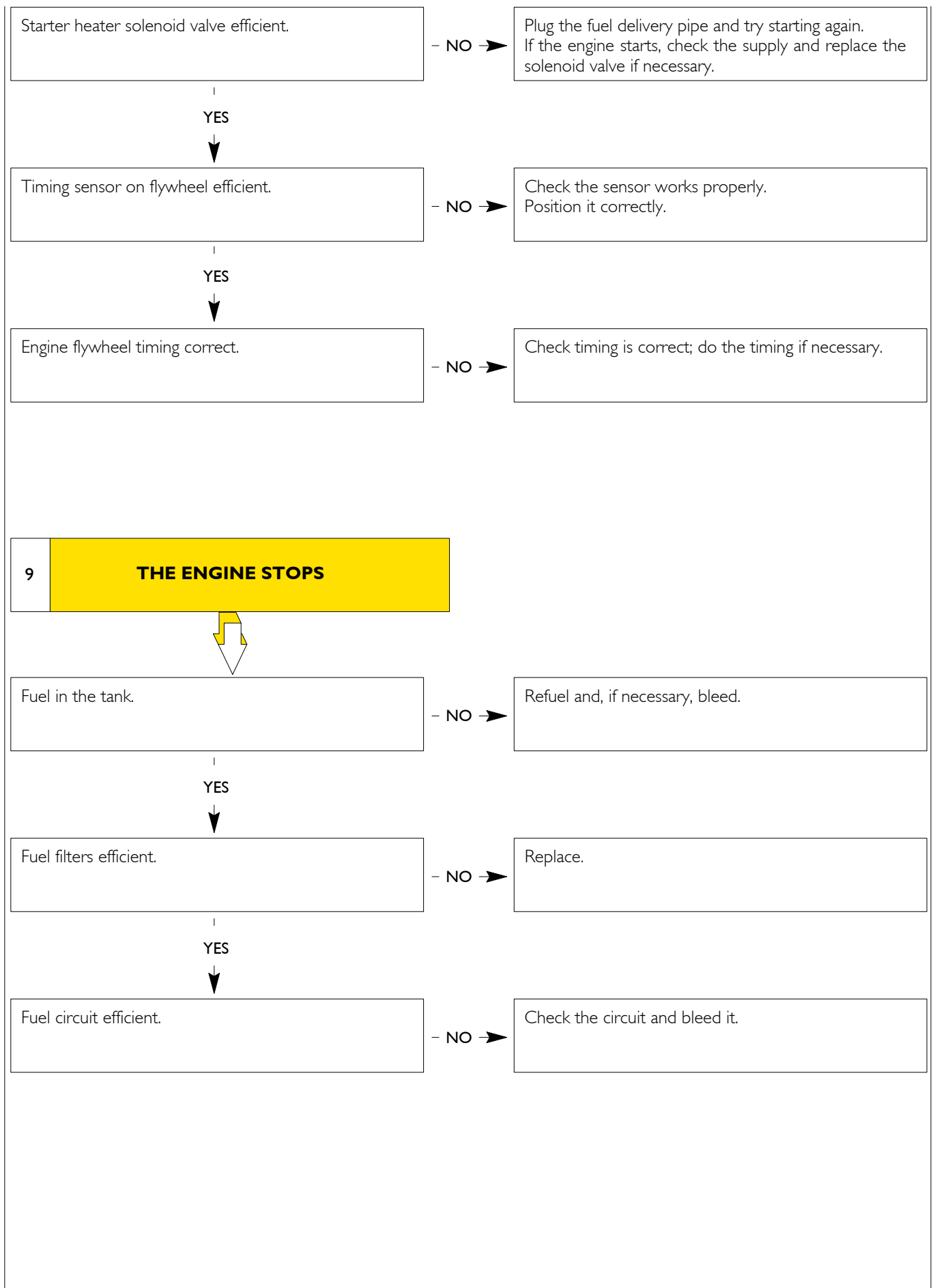
Engine flywheel timing correct.

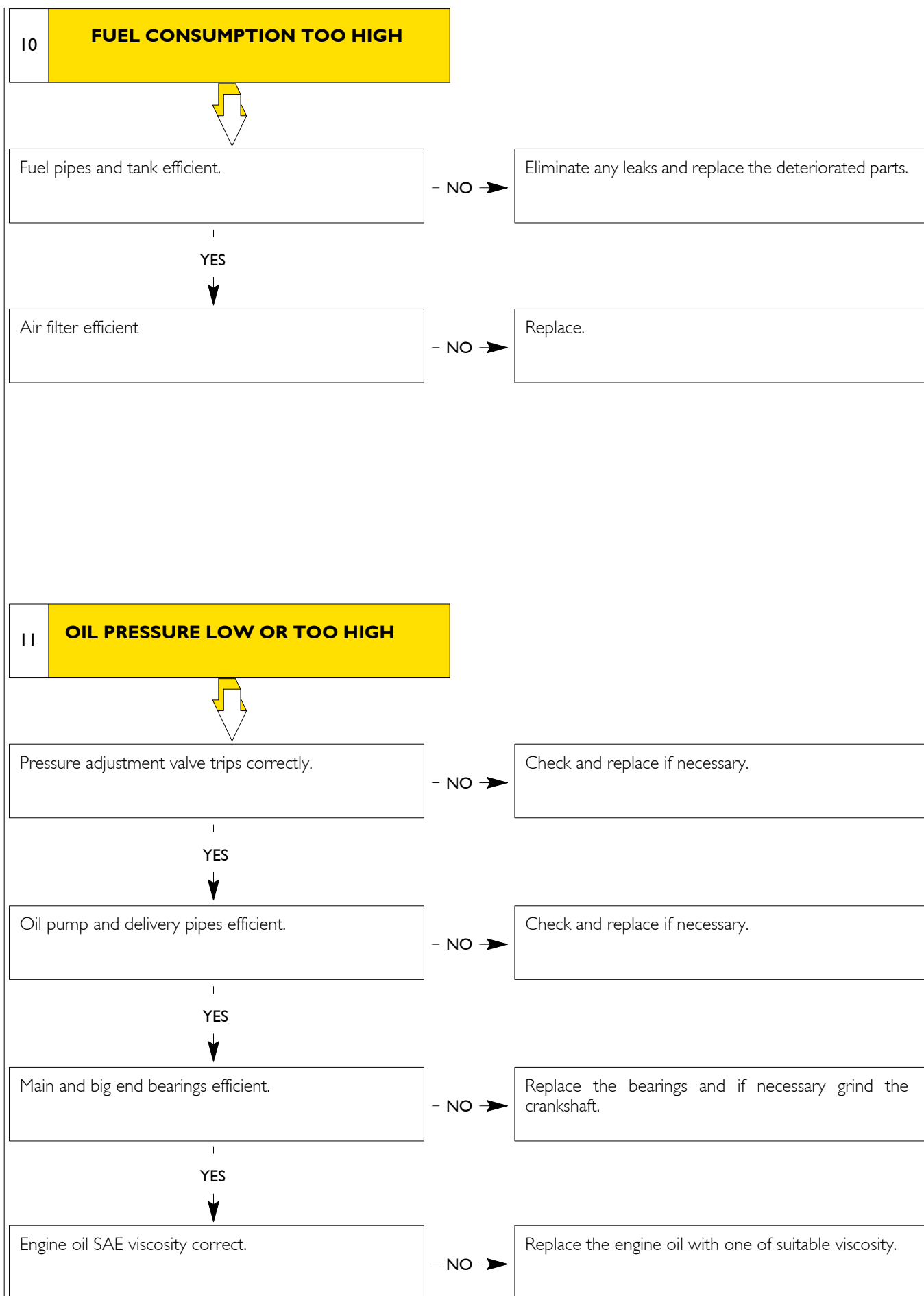
- YES →

Check the timing is correct; do the timing if necessary.



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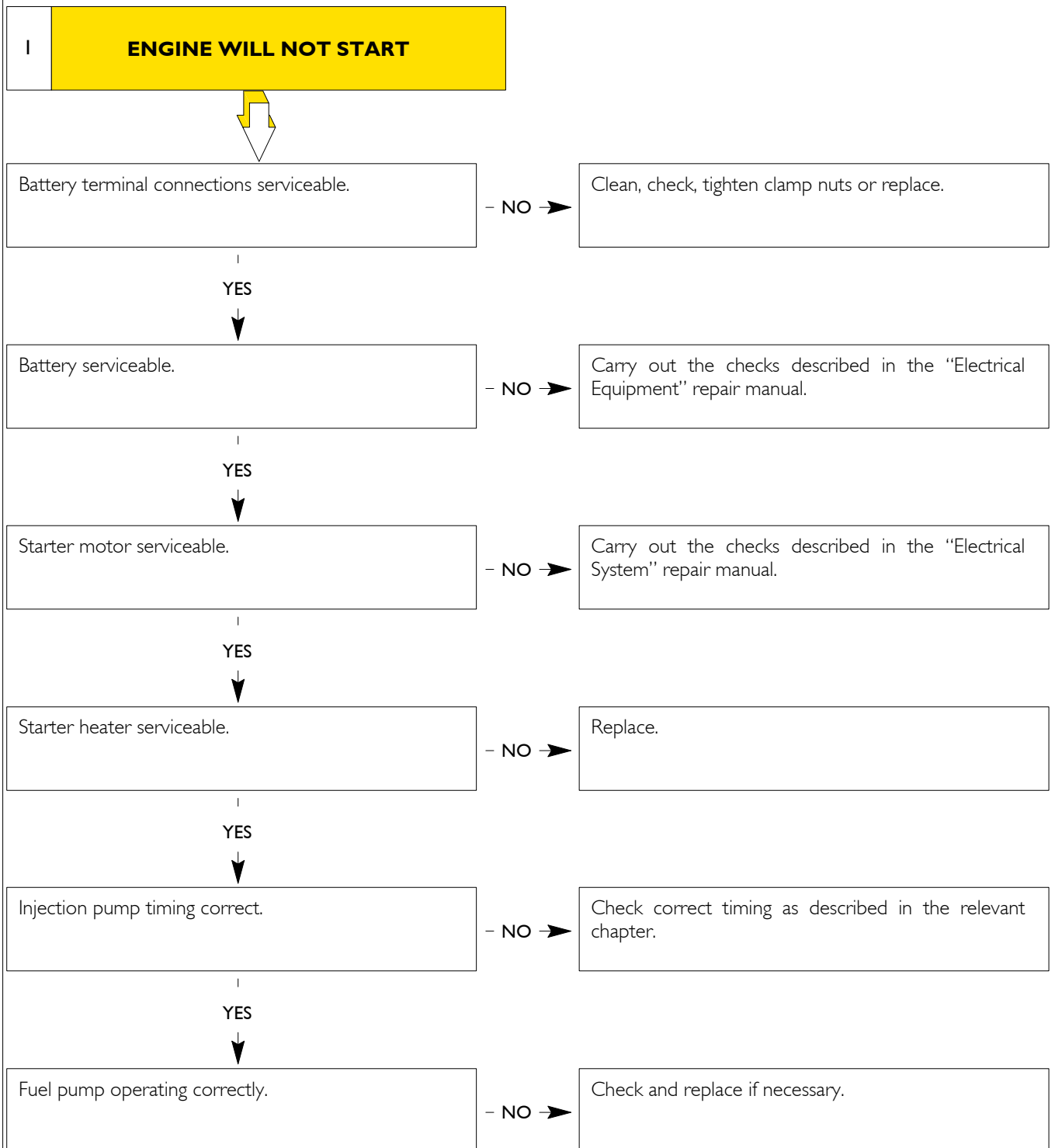
Engines with rotary mechanical injection pump

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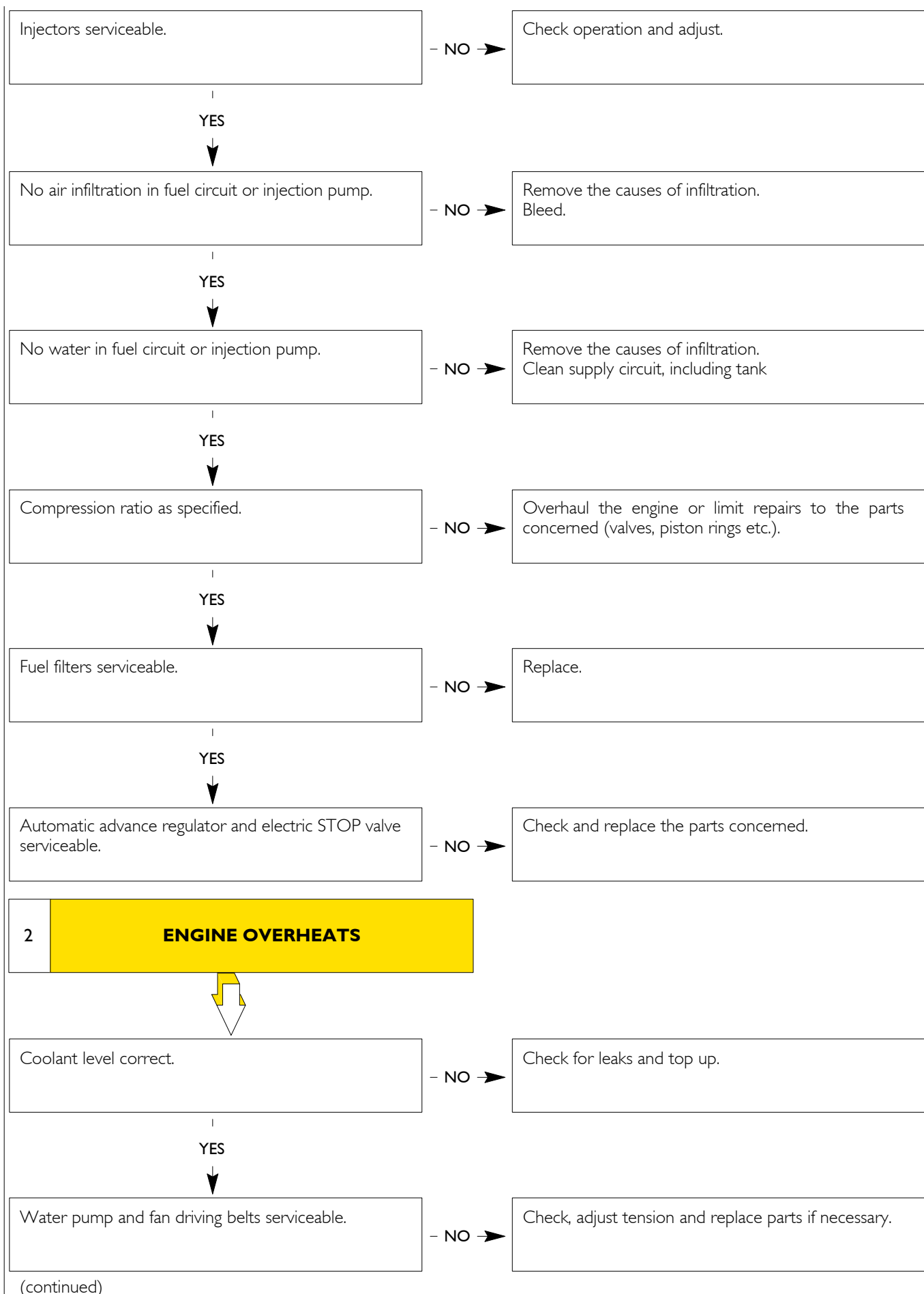
TROUBLESHOOTING

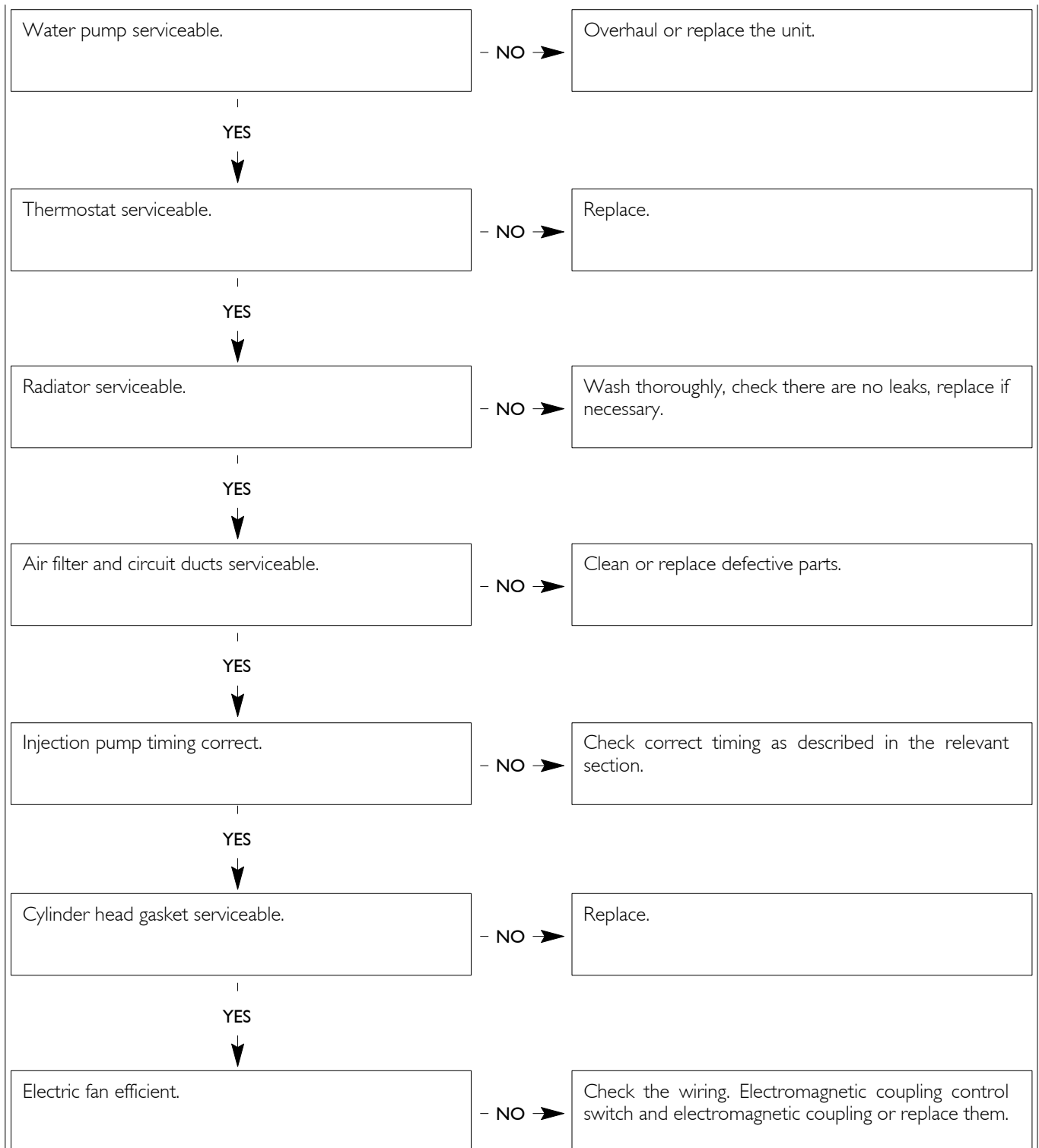
Main engine operating faults with mechanical injection pump:

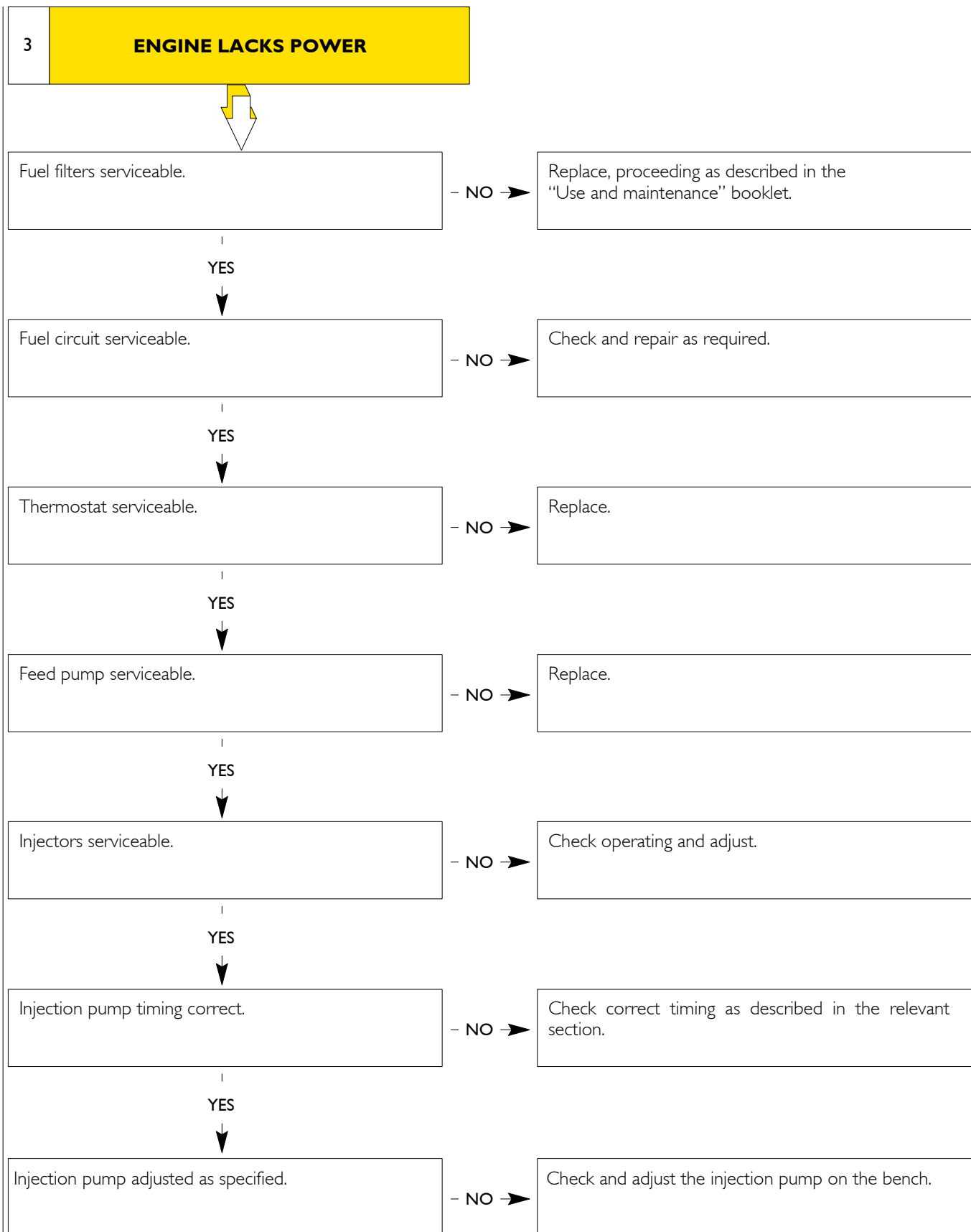
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|--|--|
| 1 - engine will not start; | 6 - engine emits blue smoke; |
| 2 - engine overheats; | 7 - abnormal knocking from the engine; |
| 3 - engine lacks power; | 8 - engine stops; |
| 4 - engine emits black or dark grey smoke; | 9 - engine exceeds maximum rpm; |
| 5 - engine emits grey (whitish) smoke; | 10 - oil pressure too high or too low; |
| | 11 - excessive fuel consumption. |



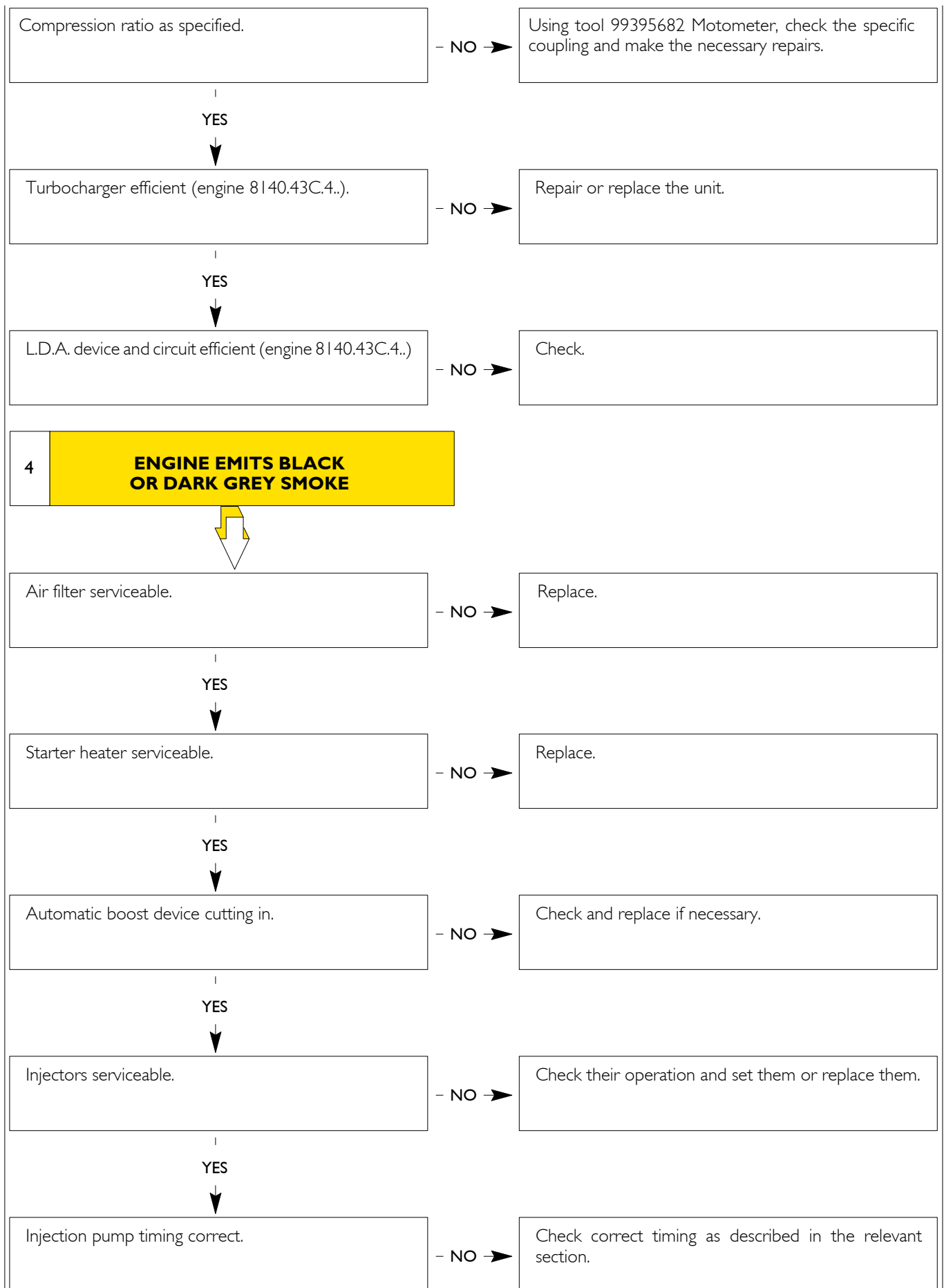
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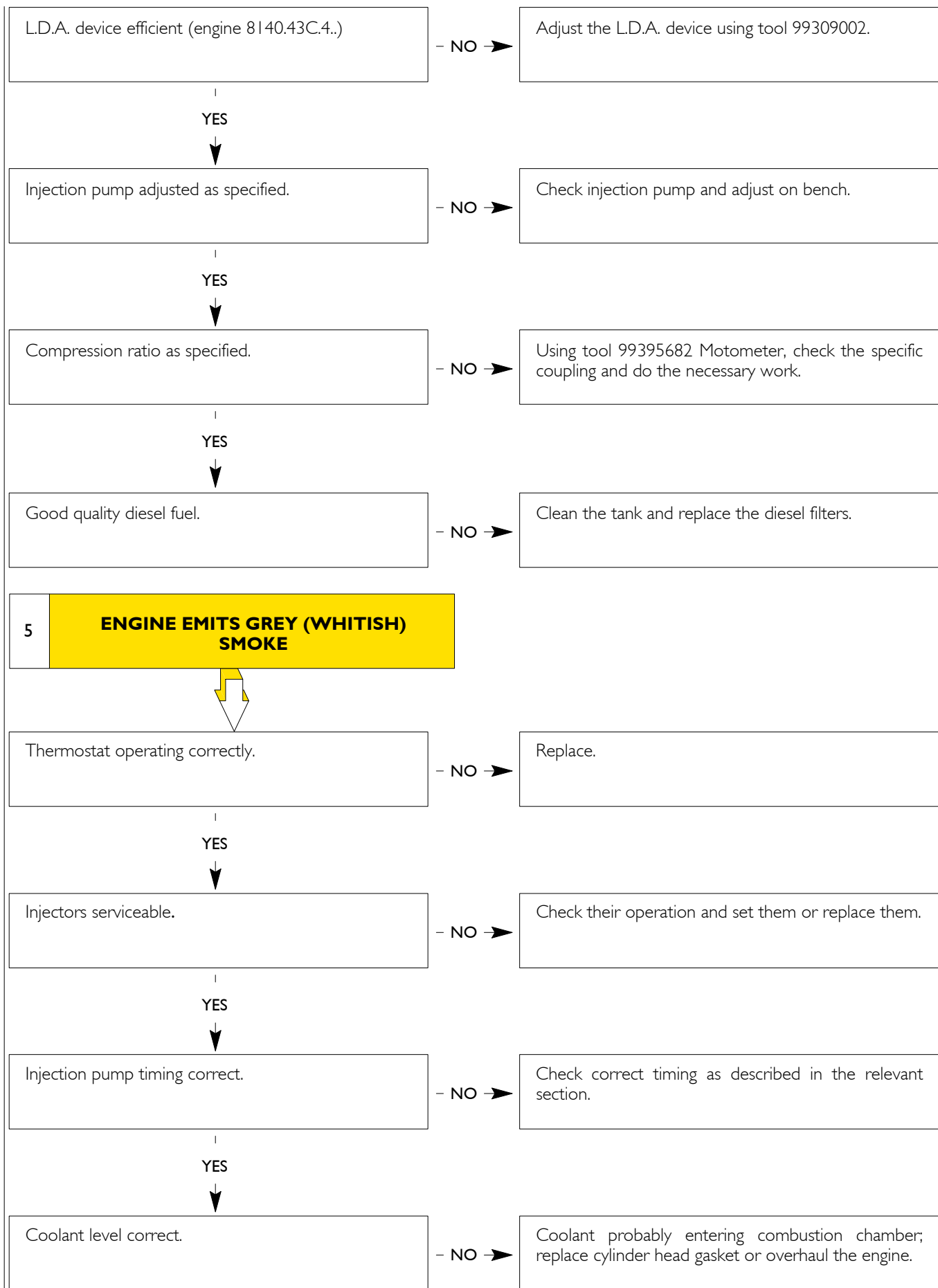




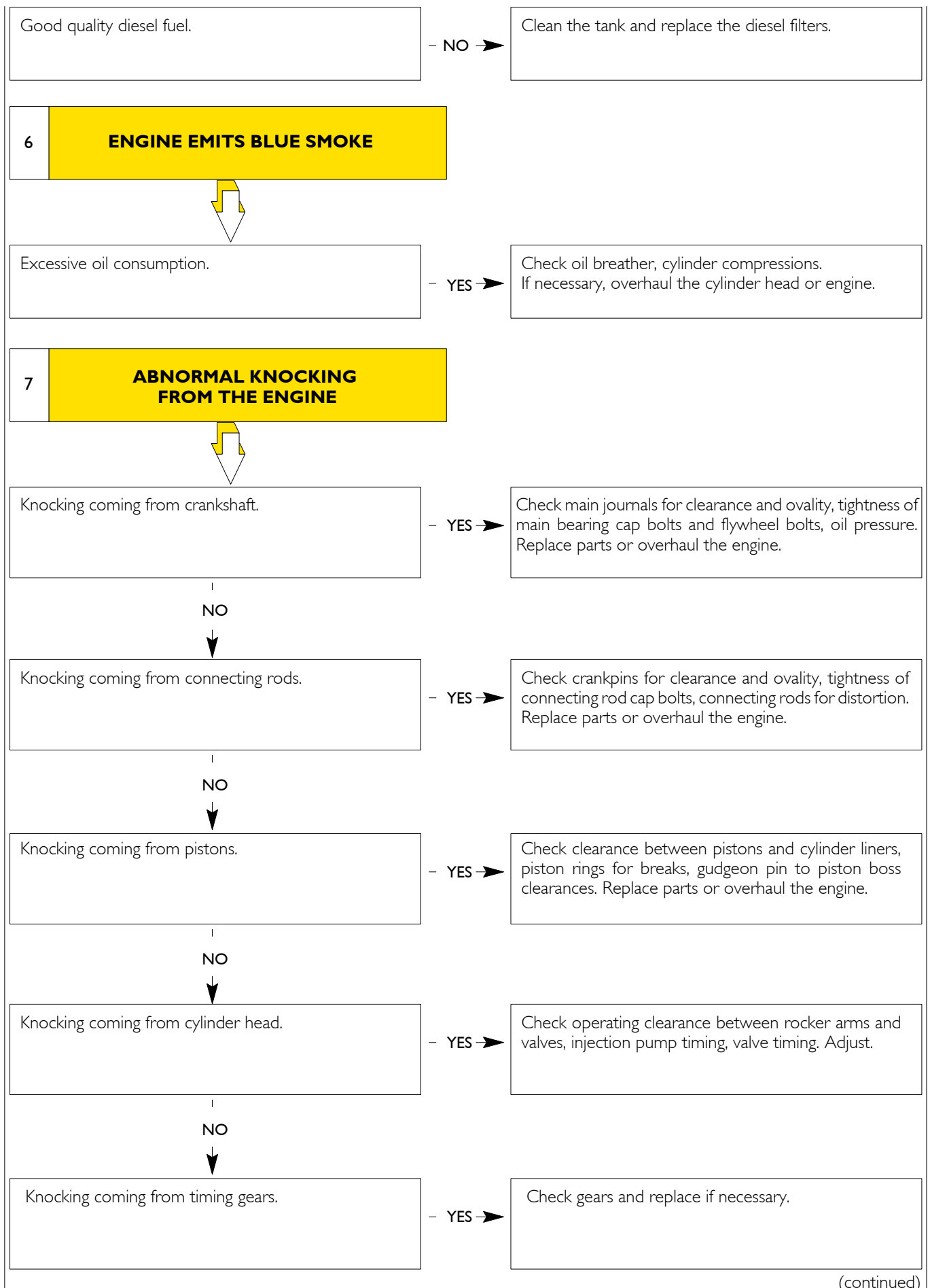
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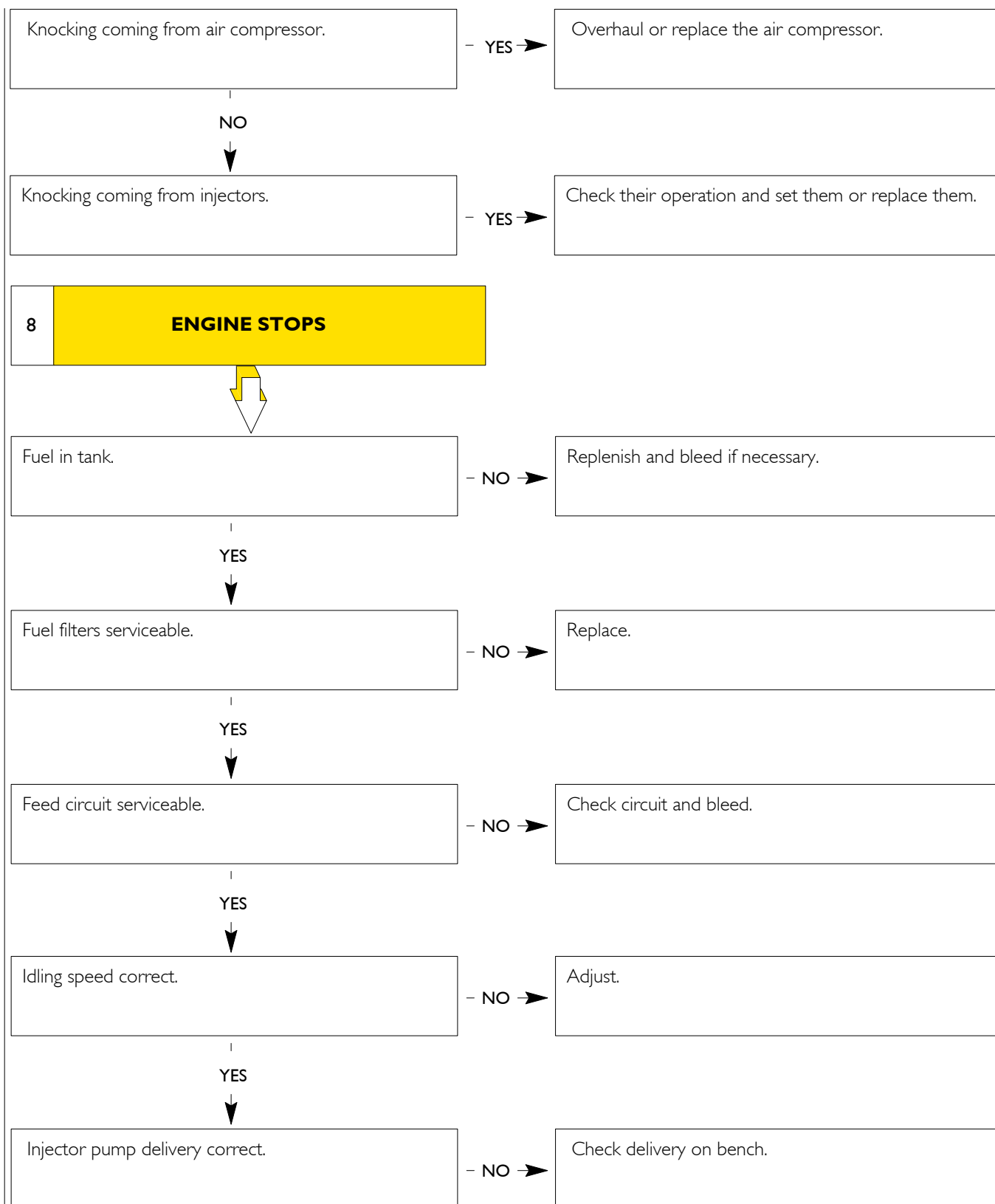
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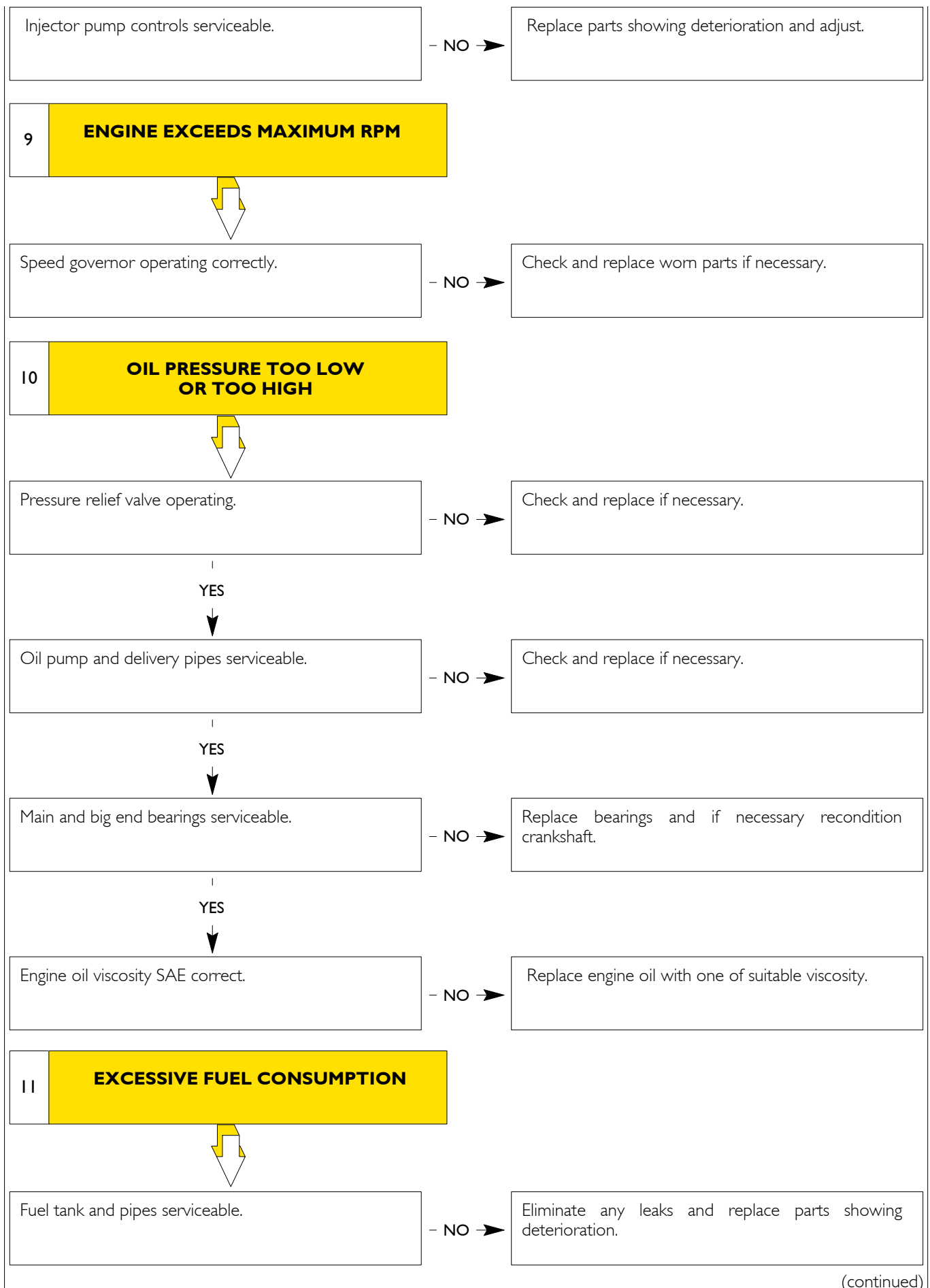
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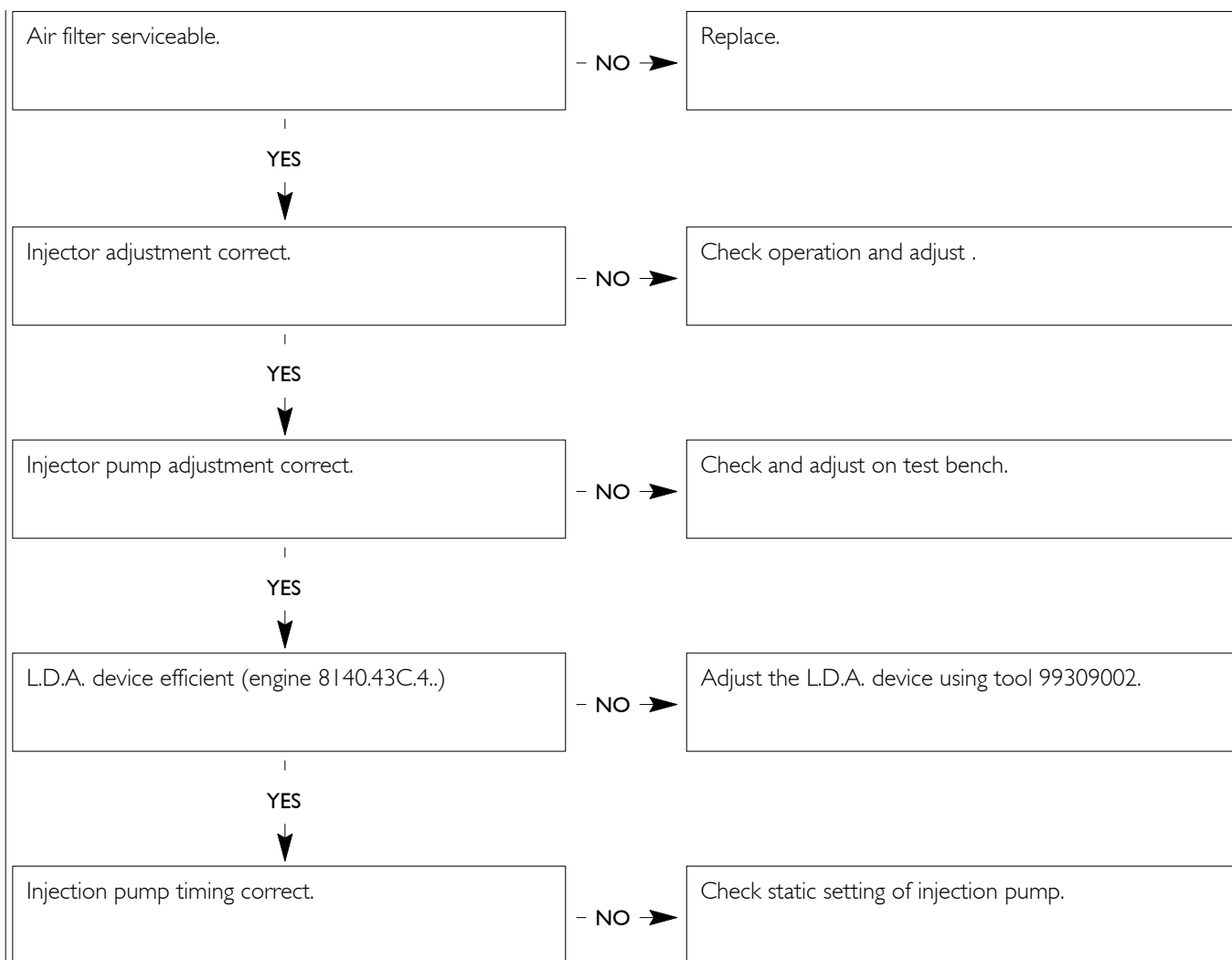
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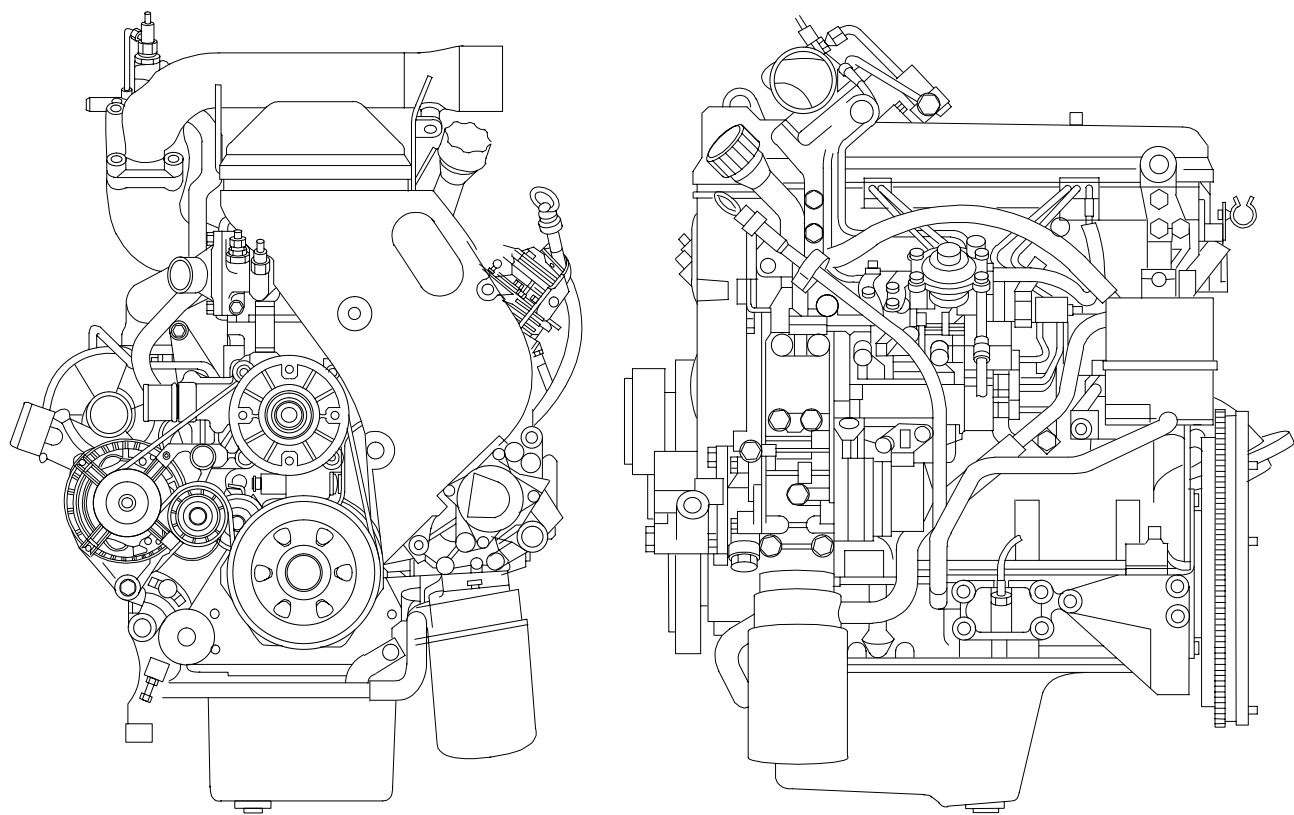
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5401 Engine 8140.43C.43XX

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Figure I



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Gas emissions

Engine 8140.43C conforms with the Euro2 standards on gas emissions (measurement on engine bench according to R 49 cycle), with the following limits fixed by EEC 91/542 standards:

- CO (carbon monoxide) < 4.0 g/kWh
- NOx (nitrogen oxide) < 7.0 g/kWh
- HC (hydrocarbons) < 1.10 g/kWh
- Particulate < 0.15 g/kWh

Test fuel: CEC RFT 73 - T - 90 - S ≤ 0.05 %.

Smokiness

Engine 8140.43C conforms with the limits of smokiness required by EEC standards 72/306 and R 24-03 with the following exhaust smokiness values:

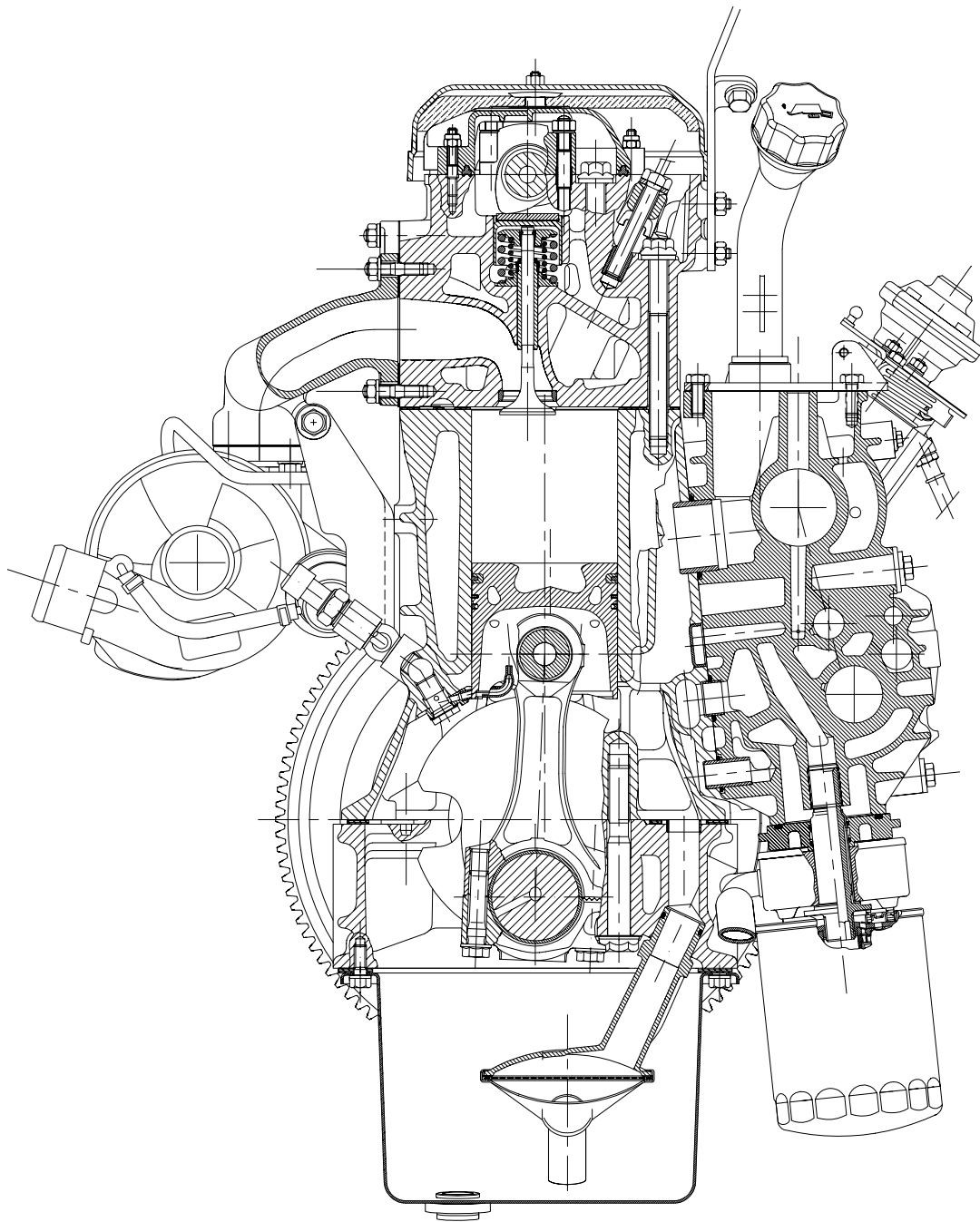
- Maximum power (Bosch BSU opacimeter degrees) 2.0
- Maximum torque (Bosch BSU opacimeter degrees) 2.5
- Full load at 1000 rpm (Bosch BSU opacimeter degrees) 3.5

Noise emissions

Maximum mean noise level, Lpa, of the standard engines measured according to ISO Std. 3745 (microphones at 1 m from the engine surfaces):

- Idling (800 rpm) dB A 79
- Full power (3600 rpm) dB A 99

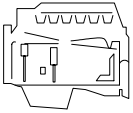
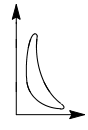
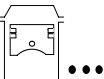
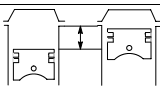
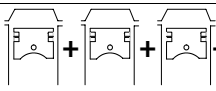
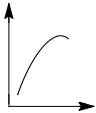
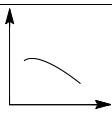
Figure 3



8140.43C.4... ENGINE CROSS SECTION

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GENERAL SPECIFICATIONS

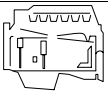
	Type	8140.43C.4.. ID/TCA	
	Cycle	Diesel 4 strokes	
	Feeding	Supercharged	
	Injection	Direct	
	No. of cylinders	4 on-line	
	Diameter	mm	94.4
	Stroke	mm	100
	Total displacement cm ³		2800
	Compression ratio	18.5	
	Max. power	kW (HP)	73.8 ÷ 78.6 100.4 ÷ 106.9
		rpm	3600
	Max. torque	Nm (kgm)	245.8 ÷ 268.6 25.1 ÷ 27.4
		rpm	1800
	Engine idling speed, no load	rpm	800 ± 25
	Maximum engine speed, no load	rpm	4200 ± 50
	Pressure at T.D.C. *bar	20 ÷ 26	
	Minimum permissible pressure at T.D.C. *bar	16	

(*) The pressure value is recorded by turning the engine over with the electric starter motor, with oil temperature at 40°- 50°C and the injection pump in the stop condition.

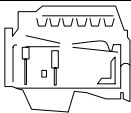
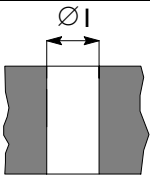
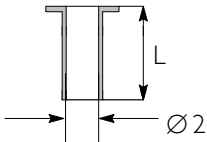
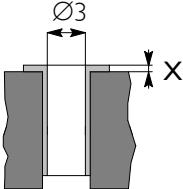
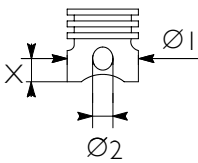
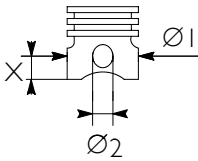
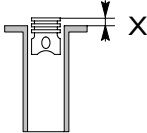
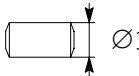
ID = Direct injection
 PC = Indirect injection (pre-chamber)
 TCA = Supercharging with intercooler

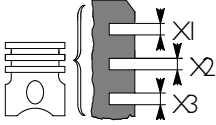
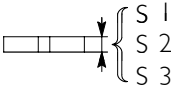
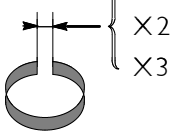
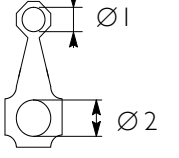
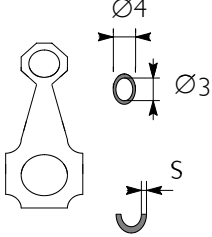
	Type	8140.43C.4.. ID/TCA
	VALVE TIMING opens before T.D.C. A closes after B.D.C. B opens before B.D.C. D closes after T.D.C. C	8° 37° 48° 8°
	For timing check X mm Running X mm X mm	0.5 ± 0.05 0.5 ± 0.05 0.5 ± 0.05 0.5 ± 0.05
	FUEL SUPPLY Bosch type injection pump - with immobilizer - without immobilizer Speed governor Cold start	rotating distributor VER 824 VER 824-I all speeds K.S.B. electrical, mechanical
	Pump arrangement With piston n. 1 at T.D.C. Start of delivery mm	1.10 ± 0.05
	BOSCH injector nozzle type	SACLESS DSLA 134 P 604
	Injection order - injection pump - engine	A - B - C - D 1 - 3 - 4 - 2
	Injection pressure bar	240 + 12 *

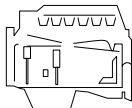
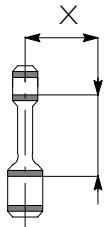
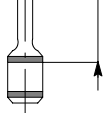
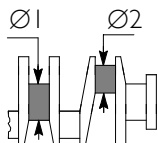
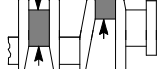
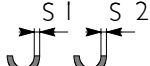
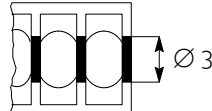
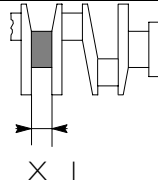
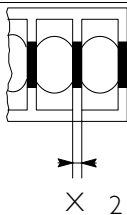
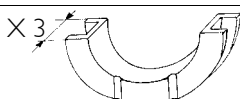
* When checking, for any values of less than 200 bars, calibrate the injectors at 230 + 8 bar.

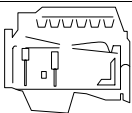
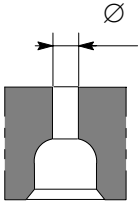
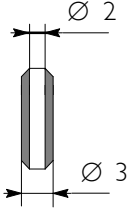
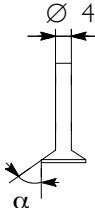
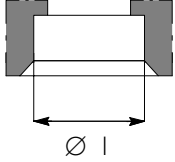
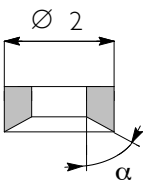
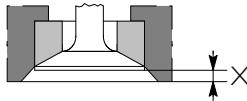
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	SUPERCHARGING Turbocharger type marked with colour	With intercooler KKK K03-2076-CCA 6.68 pink - - - 3.5 ± 0.2 - 1.3 ± 0.002
	Turbocharger type marked with colour	MITSUBISHI TD04 pink 0.396 ± 0.602 0.034 ± 0.106 1 5 1.11 ± 0.0026 1.35 ± 0.0039
	LUBRICATION Oil pressure, engine hot at idling speed at maximum speed	Forced feed by gear pump, relief valve, dual action oil filter ≥ 0.8 ≥ 3.5
COOLING		With a centrifuge pump, driven by the crankshaft via a poly-V belt, thermostat, fan with electro-magnetic clutch, radiator, heat exchanger. "Intercooler" 82° C ± 2° C 110° C
	OIL REPLENISHMENT Total capacity at 1st filling liters kg Quantity at periodical replacements: - engine sump liters kg - engine sump liters kg	6.9 6.1 5.2 4.6 6.3 5.7
Urania Daily Urania Turbo Urania LD 5		

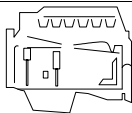
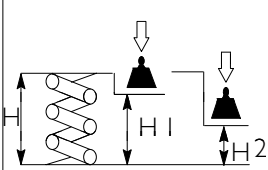
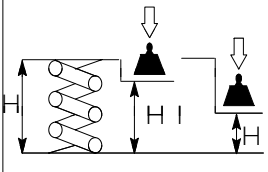
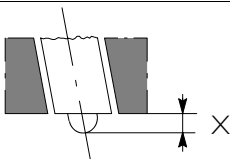
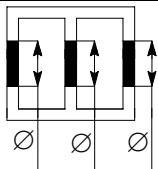
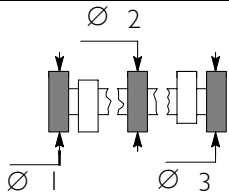
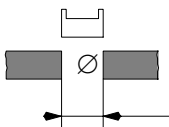
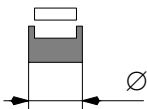
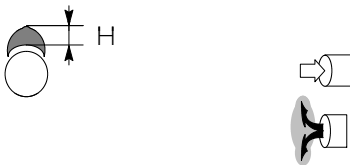
ASSEMBLY DATA - CLEARANCES

	Type	8140.43C.4.. ID/TCA	
CYLINDER BLOCK AND CRANK MECHANISM COMPONENTS			
	Bores for cylinder liners Ø 1	97.39 ÷ 97.45*	
	Cylinder liners: outside diameter Ø 2 length L	** 97.47 ÷ 97.50 167.00 ÷ 167.30	
	Cylinder liners - crankcase bores (negative allowance)	0.02 ÷ 0.11	
	Outside diameter Ø 2	0.2	
	Cylinder barrels: (protrusion from engine block bottom) inside diameter  Ø 3	** 0.005 max 94.402 ÷ 94.432	
	Pistons: supplied as spare parts type: measuring dimension X outside diameter Ø 1 pin bore Ø 2	MONDIAL PISTON 10 94.306 ÷ 94.320 32.003 ÷ 32.009	
	Piston - cylinder sleeve	0.082 ÷ 0.126	
	Pistons: supplied as spare parts type: measuring dimension X outside diameter Ø 1 pin bore Ø 2	KS 10 94.306 ÷ 94.320 32.003 ÷ 32.009	
	Piston - cylinder sleeve	0.082 ÷ 0.126	
	Piston diameter Ø 1	0.4	
	Pistons protrusion X	> 0.40 ÷ ≤ 0.80	
	Gudgeon pin Ø 3	31.990 ÷ 31.996	
	Gudgeon pin - pin housing	0.003 ÷ 0.015	
* Diameter to be obtained at servicing when fitting spare cylinder barrels.			
** Supplied as spare parts.			

	Type	8140.43C.4.. ID/TCA
CYLINDER BLOCK AND CRANK MECHANISM COMPONENTS		
	Piston type X1* Piston ring grooves X2 X3 * measured on Ø of 91.4 mm	MONDIAL PISTON - KS 2.200 ÷ 2.230 2.050 ÷ 2.070 2.540 ÷ 2.560
	S 1 S 1* S 2 Piston rings S 2 S 3 S 3 * measured on Ø of 91.4 mm	2.068 ÷ 2.097 1.970 ÷ 1.995 2.470 ÷ 2.490
	1 Piston rings - grooves 2 3	0.103 ÷ 0.162 0.55 ÷ 0.100 0.05 ÷ 0.09
		0.4
	X1 Piston ring end gap in cylinder liners X2 X3 X1 X2 X3	0.20 ÷ 0.35 0.30 ÷ 0.50 0.30 ÷ 0.55
	Small end bush housing Ø 1 Big end bearing housing * Ø 2 * spare connecting rod supplied	35.460 ÷ 35.490 60.341 ÷ 60.348
	Small end bush diameter outside Ø 4 inside Ø 3 Big end bearing shell supplied as spare parts S	35.570 ÷ 35.595 32.010 ÷ 32.020 1.875 ÷ 1.884
	Small end bush - housing	0.08 ÷ 0.135
	Piston pin - bush	0.014 ÷ 0.03
		0.254 ÷ 0.508

	Type	8140.43C.4.. ID/TCA	
CYLINDER BLOCK AND CRANK MECHANISM COMPONENTS			
	Measuring dimension X	125	
	Max. connecting rod axis misalignment tolerance =	0.07	
	Main journals n° 1 - 2 - 3 - 4 n° 5	Ø 1	80.182 ÷ 80.208 86.182 ÷ 86.208
	Crankpins	Ø 2	56.515 ÷ 56.538
	Main bearing shells	S1*	2.165 ÷ 2.174
	Big end bearing shells	S2*	1.875 ÷ 1.884
* supplied as spare parts			
	Main bearings n° 1 - 2 - 3 - 4 n° 5	Ø 3	84.588 ÷ 84.614 90.588 ÷ 90.614
	Bearing shells - main journals	0.032 ÷ 0.102	
	Bearing shells - big ends	0.035 ÷ 0.083	
	Main bearing shells	0.254 ÷ 0.508	
	Big end bearing shells	0.254 ÷ 0.508	
	Main journal, thrust bearing	X 1	31.000 ÷ 31.100
	Main bearing housing, thrust bearing	X 2	26.500 ÷ 26.550
	Thrust washer halves	X 3	30.900 ÷ 30.950
	Crankshaft end float	0.060 ÷ 0.310	

	Type	8140.43C.4.. ID/TCA	
CYLINDER HEADS - VALVE GEAR			
	Valve guide housings in the cylinder heads $\varnothing 1$	12.950 ÷ 12.985	
	Valve guide $\varnothing 2$  $\varnothing 3$	8.023 ÷ 8.038 13.012 ÷ 13.025	
	Valve guides and housings in the cylinder heads	0.027 ÷ 0.075	
	 > Valve guide	0.05 - 0.10 - 0.2	
	Valves:  $\varnothing 4$ α  $\varnothing 4$ α	7.985 ÷ 8.000 60° 15' ± 7' 30" 7.985 ÷ 8.000 45° 30' ± 7' 30"	
	Valve stem and its guide	0.023 ÷ 0.053	
	Housing in head for valve seat  $\varnothing 1$  $\varnothing 1$	42.125 ÷ 42.175	
		37.380 ÷ 37.415	
	Outside diameter of valve seat; angle of valve seat in cylinder head:  $\varnothing 2$ α  $\varnothing 2$ α	42.245 ÷ 42.260	
		60° ± 5'	
		37.495 ÷ 37.510 45° ± 5'	
	Recessing of valve  $\times$  $\times$	1.2 ÷ 1.5	
		1 ÷ 1.3	
	Between valve seat and head  	0.070 ÷ 0.135	
		0.080 ÷ 0.130	
	 > Valve seats	-	

	Type	8140.43C.4.. ID/TCA	
CYLINDER HEADS - VALVE GEAR			
	Valve outside spring height: free height H under a load of: kg 43.8 ± 2.5 H1 kg 77.4 ± 4 H2		52 38.5 28.5
	Valve inside spring height: free height H under a load of: kg 16.4 ± 1 H1 kg 30 ± 1.5 H2		45.5 33.5 23.5
	Injector protrusion X		2.23 ÷ 2.69
	Camshaft bearing housing normal Ø oversized Ø		33.985 ÷ 34.015 34.185 ÷ 34.215
	Camshaft bearing journals normal Ø oversized Ø		33.934 ÷ 33.950 34.134 ÷ 34.150
	Between seats and supporting pins		0.035 ÷ 0.081
	Tappets housing on cylinder heads Ø normal		44.000 ÷ 44.025
	Tappet Ø normal		43.950 ÷ 43.970
	Between tappets and seats		0.030 ÷ 0.075
	Cap		3.25 to 4.45 mm with a progression of 0.05
	Cam lift: H H		9.5 10.5

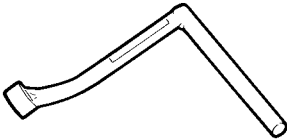
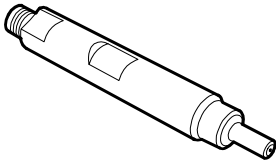
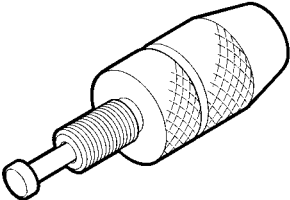
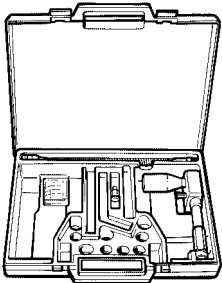
TIGHTENING TORQUE

NOTE This differs from the analogous section for 8140.43R/B/S/N engines in the following tightening torques.

PART	TORQUE	
	Nm	kgm
Nut M14 x 1.5 fixing toothed bushing to the injection pump	83	8.3
Nut or screw fixing injection pump	25	2.5
Fastening L.D.A. coupling to the intake manifold and to the injection pump	10	1
Screw fixing injector bracket	40	4
Screw fixing top cover of auxiliary element assembly	18	1.8
Fastening diesel inlet and outlet coupling on injection pump	25	2.5
Fastening diesel pipe fittings to the injection pump and injectors	33	3.3
Fastening "OUT" coupling on injection pump	25	2.5
Fixing diesel recovery pipes from the injectors	7	0.7

TOOLS

NOTE This differs from the analogous section for 8140.43R/B/S/N engines in the exclusion of tools 99317915 - 99389829 - 99389833 - 99395603 and the addition of the following tools.

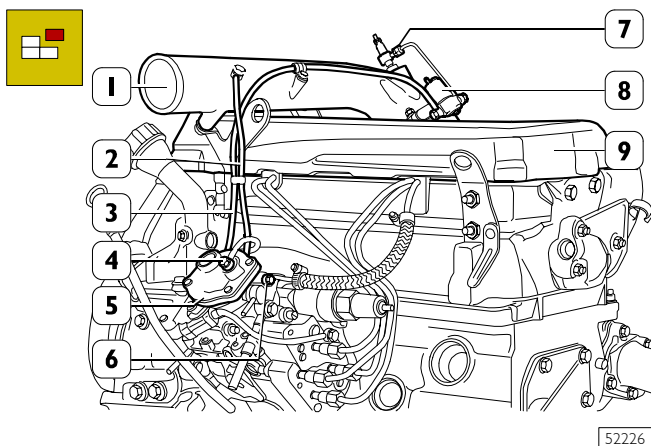
TOOL NO.	DESCRIPTION	
99352114		Wrench (13 mm) for nut on crankcase side fixing injection pump (8140.43C engines only)
99360486		Fitting to check compression in cylinders (use with 99395682)
99395100		Dial gauge holding tool for injection pump setting
99395682		Device for checking diesel engine cylinder compression

ENGINE OVERHAUL

NOTE This differs from the overhaul of the 8140.43B/R/S/N engines in the following.

540110 Dismantling engine on bench

Figure 4



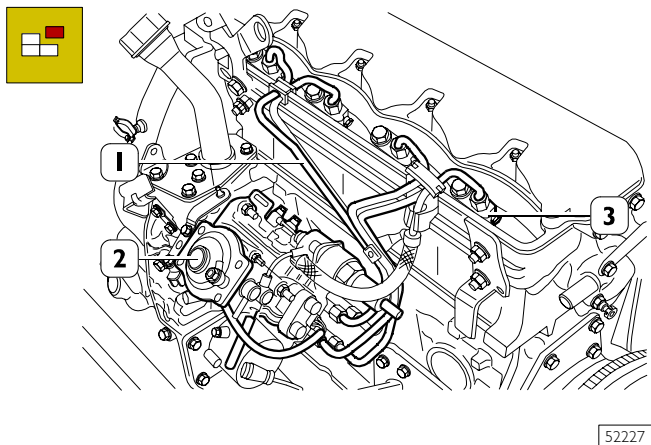
Take out the fitting (4) and disconnect the pipe (3) from the L.D.A. device (5).

Disconnect the pipe (2) from the fuel recirculation fitting (6).

Remove the air duct (1) complete with pipes (2, 3), thermostart glow plug (7) and thermostart solenoid valve (8) from the intake manifold.

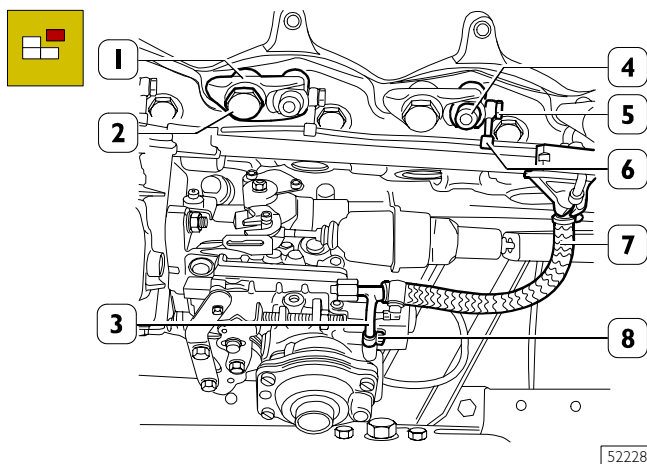
Remove the soundproofing cover (9).

Figure 5



Disconnect the fuel pipes (1) from the injectors (3) and from the injection pump (2).

Figure 6



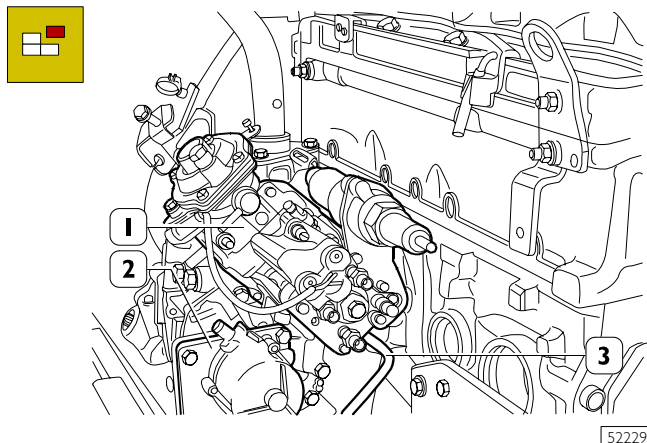
Remove connections (5) and disconnect fuel recovery pipes (6) from injectors (4).

Remove connection (8) and disconnect three-way connection (3) from injection pump including pipes (6-7) and spring plugs.

Remove injector (4) bracket (1) fixing screws (2).

Remove injectors from cylinder head.

Figure 7



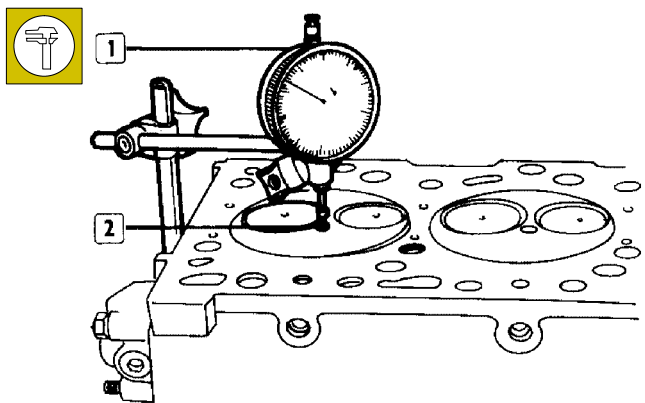
Mark the assembly position of the injection pump (1) on the auxiliary member assembly (2).

Remove the fixing nuts and disconnect the injection pump (1) from the auxiliary member assembly (2).

NOTE To remove the bottom fixing nut on the crankcase side, use wrench 993521 I 4 (3).

560610 CYLINDER HEAD

Figure 8

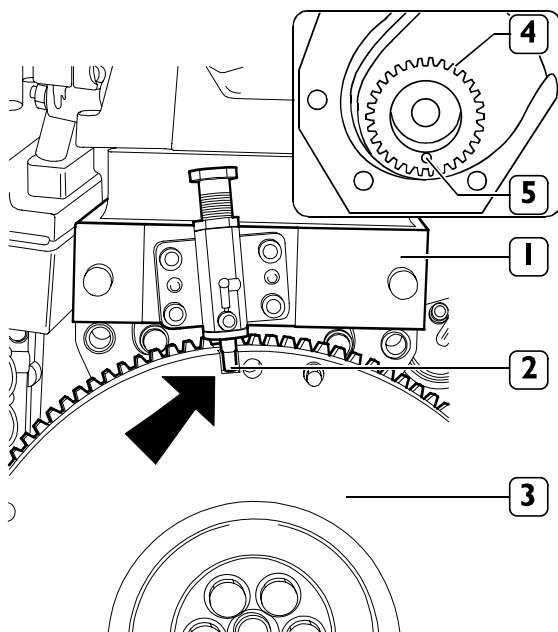


18880

Use dial gauge (1) to check whether valve sunk and injector (2) protrusion from cylinder head surface are falling within the required value.

Engine assembling at the bench Setting the timing system

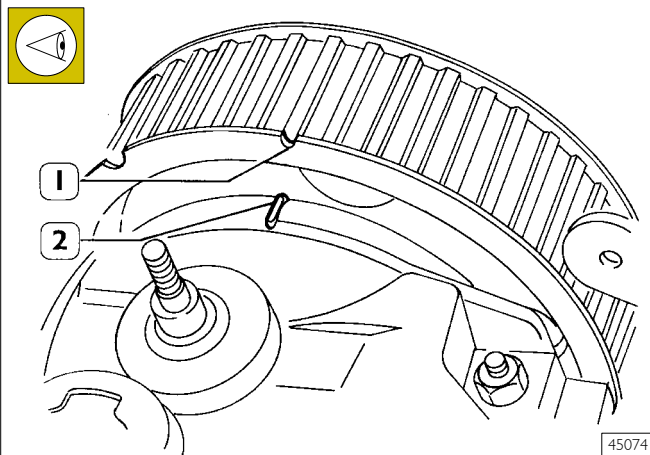
Figure 9



61900

Check whether cylinder I piston is at TDC. This condition takes place when tool 99395214 (1) pin (2) is fitted into engine flywheel (3) slit (⇒) and control gear hole (5) and timing gear (4) are facing downward.

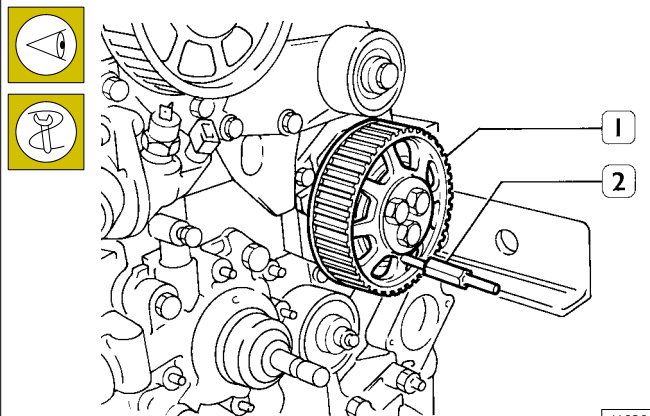
Figure 10



45074

- ☐ Make coincide notch (1) engraved on camshaft driving toothed pulley, with that (2) engraved on valve tappets cover.

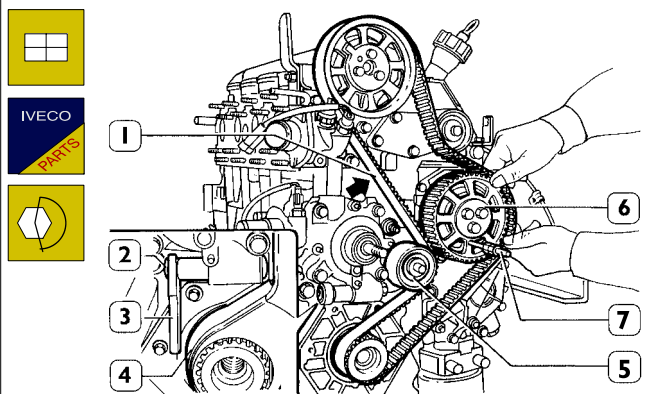
Figure 11



41820

- ☐ Rotate accessory equipments driving toothed pulley (1) so as to align pulley hole (1) with that of support and lock rotation by inserting tool 99360608 (2).

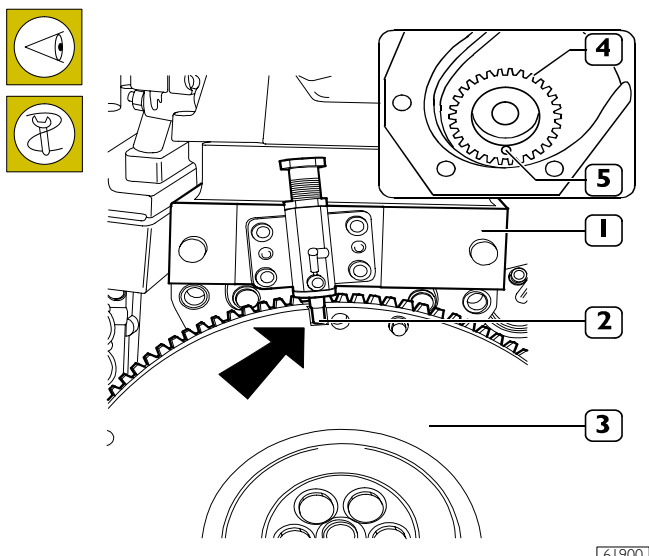
Figure 12



45079

- ☐ Fit toothed belt (1);
- ☐ fit lower cover (4) and tighten fastening screws and nuts to prescribed torque;
- ☐ remove tool (3) so as pushing rod (2), acting on belt tightener (5), makes toothed belt taut (1);

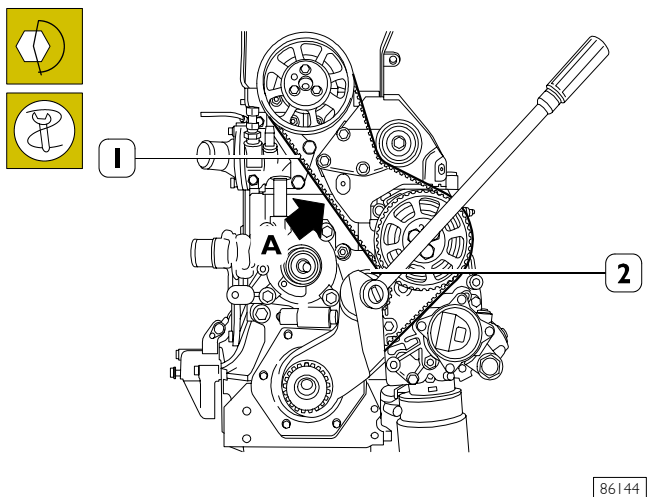
Figure 13



61900

- ☐ remove engine flywheel (3) slit ($\Rightarrow$) from tool 99395214 (1) pin (2);
- ☐ turn the crankshaft slowly in the direction of rotation by two turns and check that hole (5) in gear (4) is facing downwards and that the pin (2) of the tool (1) is inserted in the slot ($\rightarrow$) through the flywheel (3) when the timing belt is taut; the notches (1 and 2, Figure 10) must be aligned.

Figure 14



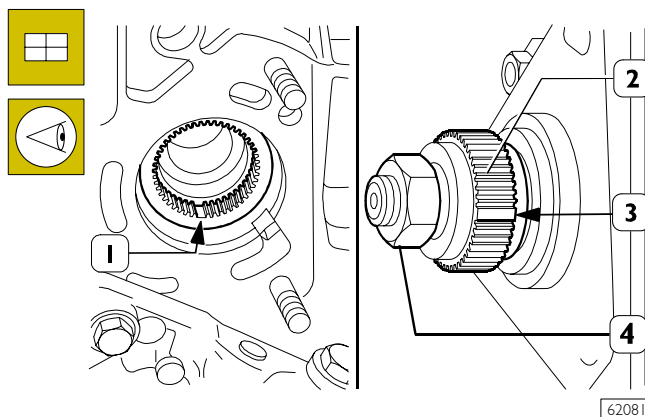
86144

- ☐ lock the tightener (2), tightening the fixing nut at a torque of 37 ± 45 Nm;

Use tool 99395849 to check timing belt tension (1) which must be 88 ± 112 Hz in point A.

Injection pump assembling and timing

Figure 15



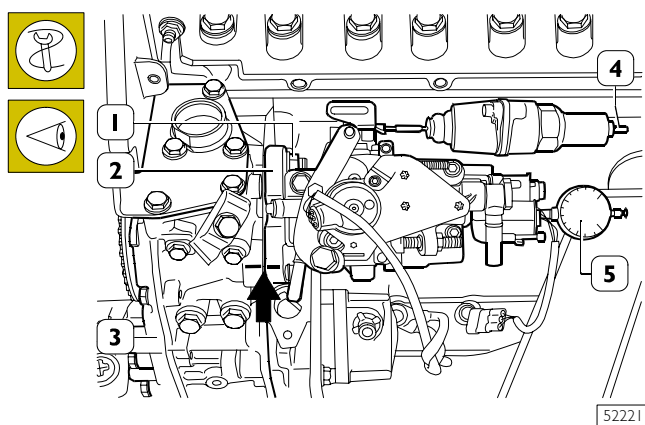
62081

NOTE Should joint (2) be replaced, use tool 99365143 to lock joint (2) rotation when moving nut (4) and tool 99342138 to remove the joint from injection pump shaft.

Connect injection pump on accessory equipments group proceeding as follows:

- ☐ check for the exact timing of valve gear;
- ☐ Fit injection pump (2, Figure 16) on auxiliary component unit (3, Figure 16). The spline (1) obtained inside pump control shaft shall coincide with joint projection (3). Position the injection pump on auxiliary component unit so that marks performed at removal coincide.
- ☐ Screw pump fastening nuts (1, Figure 16) without tightening them;

Figure 16



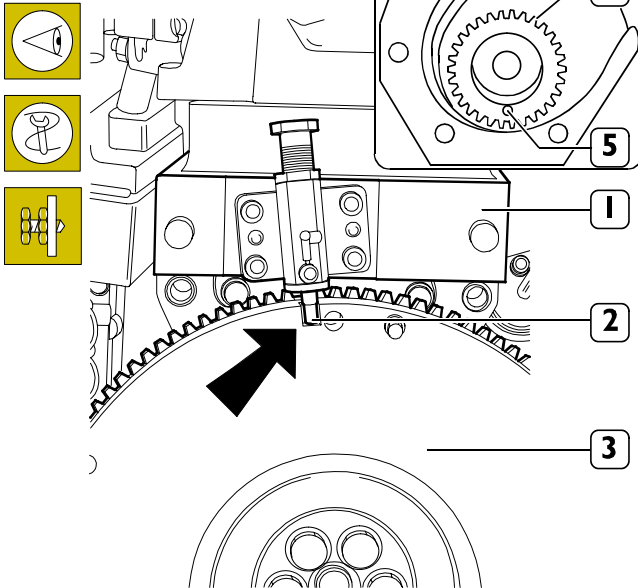
52221

- ☐ Remove the plug, situated on pump closing screw, and screw tool 99395100 (5) with the rod in contact with distributor piston crown;
- ☐ prevaricate dial gauge 99395603 (5) of ~ 3 mm;
- ☐ feed the thermal bulb (4) of K.S.B. device with a 12 V voltage, throughout timing duration; in this way K.S.B. device is cut-out.

NOTE K.S.B. device is cut-out when advance changer lever is no more engaged.

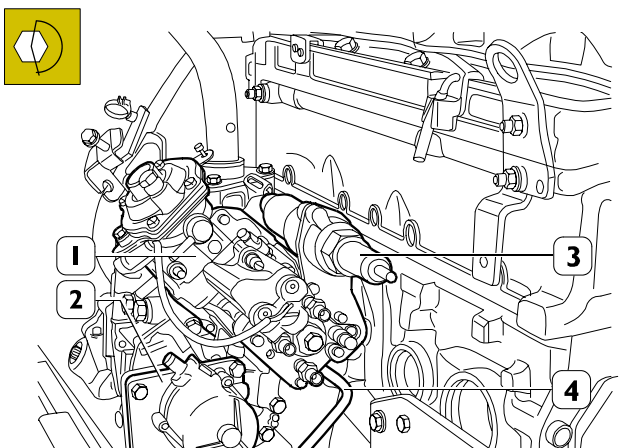
- ☐ Reverse engine rotation until dial gauge (5, Figure 16) indicates that pump piston is at BDC.
- ☐ Set dial gauge to zero.

Figure 17



- ☐ Turn the engine flywheel (3) in its direction of rotation so that, when the hole (5) of the gear (4) driving the timing system goes into its lowest position, the pin (2) of tool 99395214 (1) goes into the milled slot ($\Rightarrow$) of the engine flywheel (3).
In these conditions it is necessary to measure the stroke of the timing system piston on the dial gauge (5, Figure 16). If you find the stroke has a different value to the prescribed one, turn the pump casing in its slot to get the prescribed value indicated by the dial gauge.

Figure 18



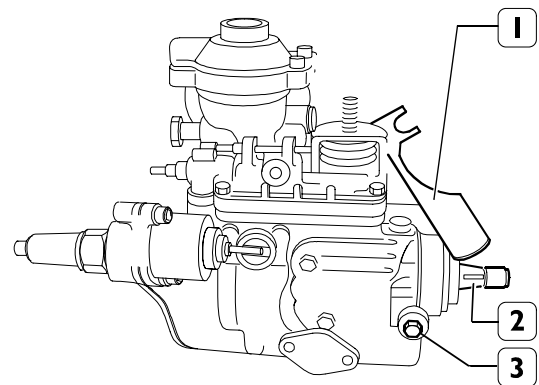
- ☐ Tighten the nuts fixing injection pump (1) to auxiliary component unit (2) to the specified torque.

NOTE To tighten injection pump fastening nut, engine crankcase side, use key 99352114 (4).

- ☐ cut-out feed to K.S.B. device (3);
- ☐ remove tool 99395100 with dial gauge (5, Figure 16) and screw again plug on locking screw;
- ☐ remove tool 99395214 (1, Figure 17).

NOTE When refitting fuel pipes, do not reuse metallic clamp with spring plug and fixing screw, but replace them with new parts.

Figure 19



NOTE The injection pump is supplied as a spare part with the start of delivery already set and with the spacer (1).

The screw (3) locks the shaft (2) of the injection pump to keep the above-mentioned setting until the pump is fitted onto the engine.

After assembly, loosen the screw (3), to make it possible to insert the spacer (1) between the screw and the casing of the injection pump, then lock the screw (3), the spacer (1) is used to ensure the shaft of the injection pump turns.

After tightening the screw (3), twist the end of the spacer (1) so as to cause it to break off and then be removed.

After fitting the injection pump, check its timing as described under the relevant heading.

5432 COOLING

Coolant temperature:

- ☐ turning on the fan: $92 \pm 2^\circ\text{C}$;
- ☐ turning off the fan: $87 \pm 2^\circ\text{C}$.

Feeding

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☐ Description	247
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NOTE The test and adjustment values indicated in the text and summary tables are given as a general indication.

INJECTION PUMP TEST VALUES

IVECO ENGINEERING		No. 500347285	Edit. I
INJECTION ROTARY PUMP TEST VALUES			
Injection pump: VE4/12FI800R824 Speed governor: All speeds		IVECO PART N.: 500329477 BOSCH PART N.: 0460 424 177	
Engine : 8140.43C.4000		For: IVECO	
Timing: - Injection pump static advance (Output reference "A") 1.0 ± 0.5 degrees before X after T.D.C. with I.P. pumping in phase of delivery at 1 mm from B.D.C. - Or with piston at T.D.C. and injection pump pumping in phase of delivery at 1.10 ± 0.04 mm from B.D.C. - Engine idling speed without load: 800 ± 25 rpm			
TEST BENCH WITH STATIC AND DYNAMIC CHARACTERISTICS CONFORMING TO STANDARDS ISO 4008/1 -/2			
- Rotation direction: Right X Left		- Supply pressure: 0.35 ± 0.05 bar	
- Pipes: Ø 6 x Ø 2 x450 mm		- Burette evacuation time:	
- Injectors: 1688 901 027		- Test fluid conforming to standard: ISO 4113	
- Calibrated hole pad: = Ø 0.5 mm		- Temperature Influx X Defluxion	
- Injector setting : 250 ± 3 bar		- Thermometer : 55 ± 1 °C	
- Backflow chock: Ø 0.75 mm		- Electronic : ± 4 °C	
"INJECTION PUMP ADJUSTMENT VALUES" "THE NUMBERS GIVEN IN BRACKETS SHOULD BE USED ONLY AS CONTROL VALUES"			
1 - START OF DELIVERY		5 - DELIVERY AT FULL LOAD WITHOUT OVERCHARGING	
Pre-lift (from B.D.C.) mm: 0 ± 0.02		Speed rpm: 550	
2 - ADVANCE CHANGER TRAVEL		Delivery per 1000 shots cm³: 28.5 ± 0.5	
Speed rpm: 900		Max difference cm³:	
Overcharging pressure hPa: 1200		6 - IDLE SPEED ADJUSTMENT	
Adjustment values mm: 1.2 ± 0.1		Speed rpm: 350	
3 - TRANSFER PUMP PRESSURE		Delivery per 1000 shots cm³: 7.0 ± 2.0	
Speed rpm: 900		Max difference cm³: 6.0 (6.5)	
Overcharging pressure hPa: 1200		7 - ADJUSTING THE RESIDUAL DELIVERY	
Adjustment values bar: 5.9 ± 0.3		Speed rpm:	
4 - DELIVERY AT FULL LOAD WITH OVERCHARGING		Delivery per 1000 shots cm³:	
Speed rpm: 1100		8 - MAXIMUM SPEED ADJUSTMENT	
Overcharging pressure hPa: 1200		Speed rpm: 2000	
Delivery per 1000 shots cm³: 57.5 ± 0.5		Overcharging pressure hPa: 1200	
Max difference cm³: 4.0 (4.5)		Delivery per 1000 shots cm³: 43.0 ± 0.5	

Date: 17-11-1998

Sheet 1/4

IVECO ENGINEERING				No. 500347285		Edit. I	
INJECTION ROTARY PUMP TEST VALUES							
9 - START				14 - DELIVERY RUN AND ADJUSTMENT			
Speed	rpm:	100		.1 Speed	rpm:	500 (*)	
Delivery per 1000 shots	cm ³ :	75.0 ± 20		Overcharging pressure	hPa:	500	
Minimum delivery	cm ³ :	55.0		Delivery per 1000 shots	cm ³ :	48.5 ± 0.5 (± 3.5)	
10 - "LFB" SETTING							
Speed	rpm:			.2 Speed	rpm:	2230	
Overcharging pressure	hPa:			Overcharging pressure	hPa:	1200	
.1. Delivery difference	cm ³ :			Delivery per 1000 shots	cm ³ :	1.5 ± 1.5 (± 2.0)	
per 1000 shots							
.2. Transfer pump pressure	bar:			.3 Speed	rpm:	2100	
difference				Overcharging pressure	hPa:	1200	
11 - INJECTION ADVANCE RUN				Delivery per 1000 shots	cm ³ :	21.0 ± 6.0 (± 8.0)	
Overcharging pressure	hPa:	1200					
.1 Speed	rpm:	900		.4 Speed	rpm:	2000	
Advance travel	mm:	1.2 ± 0.1		Overcharging pressure	hPa:	1200	
	mm:	(± 0.7)		Delivery per 1000 shots	cm ³ :	43.0 ± 0.5 (± 4.5)	
.2 Speed	rpm:	1500					
Advance travel	mm:	3.6 ± 0.6		.5 Speed	rpm:	1800	
	mm:	(± 0.7)		Overcharging pressure	hPa:	1200	
.3 Speed	rpm:			Delivery per 1000 shots	cm ³ :	5.8 ± 2.5 (± 3.5)	
Advance travel	mm:						
.4 Speed	rpm:			.6 Speed	rpm:	1100	
Advance travel	mm:			Overcharging pressure	hPa:	1200	
	mm:			Delivery per 1000 shots	cm ³ :	57.5 ± 0.5 (± 3.5)	
12 - TRANSFER PRESSURE RUN							
Overcharging pressure	hPa:	1200		.7 Speed	rpm:	600	
.1 Speed	rpm:	900		Overcharging pressure	hPa:	1200	
Transfer pump pressure	bar:	5.9 ± 0.3		Delivery per 1000 shots	cm ³ :	60.5 ± 4.0 (± 6.0)	
.2 Speed	rpm:	1800					
Transfer pump pressure	bar:	8.3 ± 0.5		.8 Speed	rpm:	550	
.3 Speed	rpm:	600		Overcharging pressure	hPa:	0	
Transfer pump pressure	bar:	4.9 ± 0.5		Delivery per 1000 shots	cm ³ :	28.5 ± 0.5 (± 3.5)	
.4 Speed	rpm:						
Transfer pump pressure	bar:			.9 Speed	rpm:		
13 - BACKFLOW VOLUME ON VALVE				Overcharging pressure	hPa:		
.1 Speed	rpm:	600		Delivery per 1000 shots	cm ³ :		
Overcharging pressure	hPa:	1200					
Backflow volume	c.c./10s:	113 ÷ 146		.10 Speed	rpm:		
	c.c./10s:	(107 ÷ 152)		Overcharging pressure	hPa:		
.2 Speed	rpm:	1800		Delivery per 1000 shots	cm ³ :		
Overcharging pressure	hPa:	1200					
Backflow volume	c.c./10s:	143 ÷ 187					
	c.c./10s:	(138 ÷ 193)					

Date: 17-11-1998

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IVECO ENGINEERING				No. 500347285		Edit. I	
INJECTION ROTARY PUMP TEST VALUES							
15 -DELIVERY NULL (STOP)				19 - POTENTIOMETER CALIBRATION AND CHECKING			
1	Mechanical:			Supply	volt:		
	Speed	rpm:	425	Setting			
	Delivery per 1000 shots	cm ³ :	1.5 ± 1.5	Speed	rpm:		
	Voltage	volt:	12	Delivery per 1000 shots	cm ³ :		
2	Electrical:			Potentiometer output voltage	volt:		
	Speed	rpm:	350	Checking:			
	Delivery per 1000 shots	cm ³ :	1.5 ± 1.5	Speed	rpm:		
	Voltage	volt:	0	Delivery per 1000 shots	cm ³ :		
16 - DELIVERY AT IDLING SPEED				Potentiometer output voltage	volt:		
1	Speed	rpm:	350	20 - EGR MICROSWITCH CALIBRATION			
	Delivery per 1000 shots	cm ³ :	7.0 ± 2.0 (± 4.0)	Speed	rpm:		
2	Speed	rpm:		Overcharging pressure	hPa:		
	Delivery per 1000 shots	cm ³ :		Delivery per 1000 shots	cm ³ :		
3	Speed	rpm:		21 - GLOW PLUG MICROSWITCH ADJUSTMENT			
	Delivery per 1000 shots	cm ³ :		Contact closed at:			
4	Speed	rpm:		control lever travel			
	Delivery per 1000 shots	cm ³ :		of injection pump			
	Residual delivery check			of injection			
5	Speed	rpm:		22 - CUT-OUT ELECTROMAGNET			
	Delivery per 1000 shots	cm ³ :		Operating minimum voltage	volt:	10	
17 - "LFB" CHECK				Rated voltage	volt:	12	
	Speed	rpm:		23 - ASSEMBLING AND ADJUSTING DIMENSIONS			
	Overcharging pressure	hPa:		Denomination:			
1	Delivery difference	cm ³ :		K	mm:	3.6	
	per 1000 shots			MS	mm:	0.9	
	Advance travel diff.	mm:		Ya	mm:	34.0 ± 1.0	
2	Delivery per 1000 shots	cm ³ :		Yb	mm:	47.0 ± 3.0	
	Advance travel diff.	mm:					
18 - AUTOMATIC START EXTRA DELIVERY				Devices			
1	Speed	rpm:	100	LDA STROKE	mm:	5.2 + 0.2	
	Delivery per 1000 shots	cm ³ :	75.0 ± 20.0	KSB LEVER	:		
		cm ³ :	min. 55	KSB ADVANCE	mm:	5.71	
2	Speed	rpm:	370	KSB ADVANCE A.P. DEGREES	:	2.0	
	Delivery per 1000 shots	cm ³ :	55.0 ± 20.0				
		cm ³ :	max 75				
3	Speed	rpm:					
	Delivery per 1000 shots	cm ³ :					
		cm ³ :					

Date: 17-11-1998

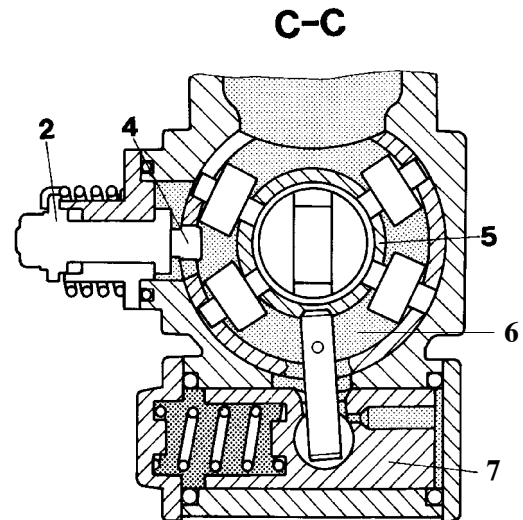
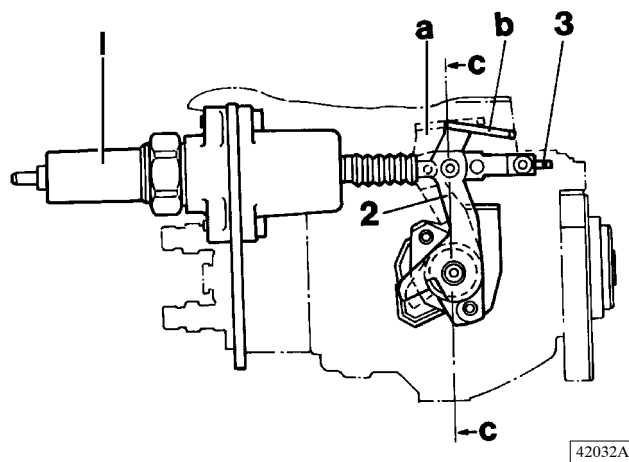
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IVECO ENGINEERING	No.500347285	Edit. I
INJECTION ROTARY PUMP TEST VALUES		
REMARKS: 1) References for which no letters or numbers are given are reserved for other injection pump designs. 2) For a correct adjustment of injection pumps on the electric bench, conform to the manual IVECO Technical Publications and Training No. XXXXXXXXXX 3) Measurements taken on injection pumps equipped with "KSB" must be performed with the same at rest. 4) (*) LDA adjusting point: act on spring pre-loading ring. 5) After every variation of LDA pressure, act on the control lever. 6) For adjusting the starting point (ARF valve) add a 12.0 mm spacer to the third delivery stop.		
NOTES: BOSCH SHEET ED. 3 DTD. 10.07.98	C . I . D . I-II4734-006-085	DATE 17-11-1998

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COLD INJECTION ADVANCE AUTOMATIC DEVICE (KSB) [KALT START BETRIEBSEINRICHTUNG]

Figure 20



1. Wax thermal bulb - 2. Control lever - 3. Flexible connecting cable - 4. Eccentric pin - 5. Roll-carrier ring - 6. Advance changer piston - a. Lever position (2) in cold start condition - b. Lever position (2) in normal operating condition.

Description

Rotary injection pumps, with mechanical speed governor, that are fitted on the engines dealt with in this publication, are equipped with a cold injection advance automatic device (KSB). This tool has the purpose of setting the injection advance, below a given temperature, to a greater value than that static one of pump fitting to the engine, and to reset it by degrees to the value of this latter, as temperature rises.

Operation

KSB cold injection advance automatic device is a mechanical type electric drive tool that mainly consists of a wax thermal bulb (1) that acts on the advance changer control lever (2), through a pushing rod and a connecting cable (3).

When starting a cold engine, the thermal bulb (1) pushing rod, that operates on the connecting cable (3), is placed at back position; consequently the lever (2) is kept in position "a".

In this condition the eccentric pin (4) (which is operated by the lever (2)), changes the position of roll carrier ring (5) with respect to its normal position and, in its turn, it causes piston (6) to move, thus producing an increase of advance with respect to the value of fitting static advance.

At engine start the resistance existing in the thermal bulb (1) is supplied by alternator D⁺.

The heat generated by the flow of current, heats the wax that expands and causes pushing rod to come out, resetting the lever (2) to position "b" by degrees.

In this way the extra advance is cancelled, since the parts of advance changer come back to their normal operating position.

The full cut-out of KSB device is achieved after about 3-4 minutes of operation.

NOTE The check and adjustment, if required, of KSB tool is carried out with the injection pump placed on a test bench.

Coolant temperature values for KSB heat bulb connection

Engine 8140.43C:

- ☐ connection at $45^{\circ}\text{C} \pm 2^{\circ}\text{C}$
- ☐ disconnection at $30^{\circ}\text{C} \pm 2^{\circ}\text{C}$

Engines 8140.63 - 8140.43S

- ☐ at altitudes < 1000m:
 - connection at $25^{\circ}\text{C} \pm 3^{\circ}\text{C}$
 - disconnection at $20^{\circ}\text{C} \pm 3^{\circ}\text{C}$
- ☐ at altitudes > 1000m:
 - connection at $55^{\circ}\text{C} \pm 4^{\circ}\text{C}$
 - disconnection at $50^{\circ}\text{C} \pm 4^{\circ}\text{C}$

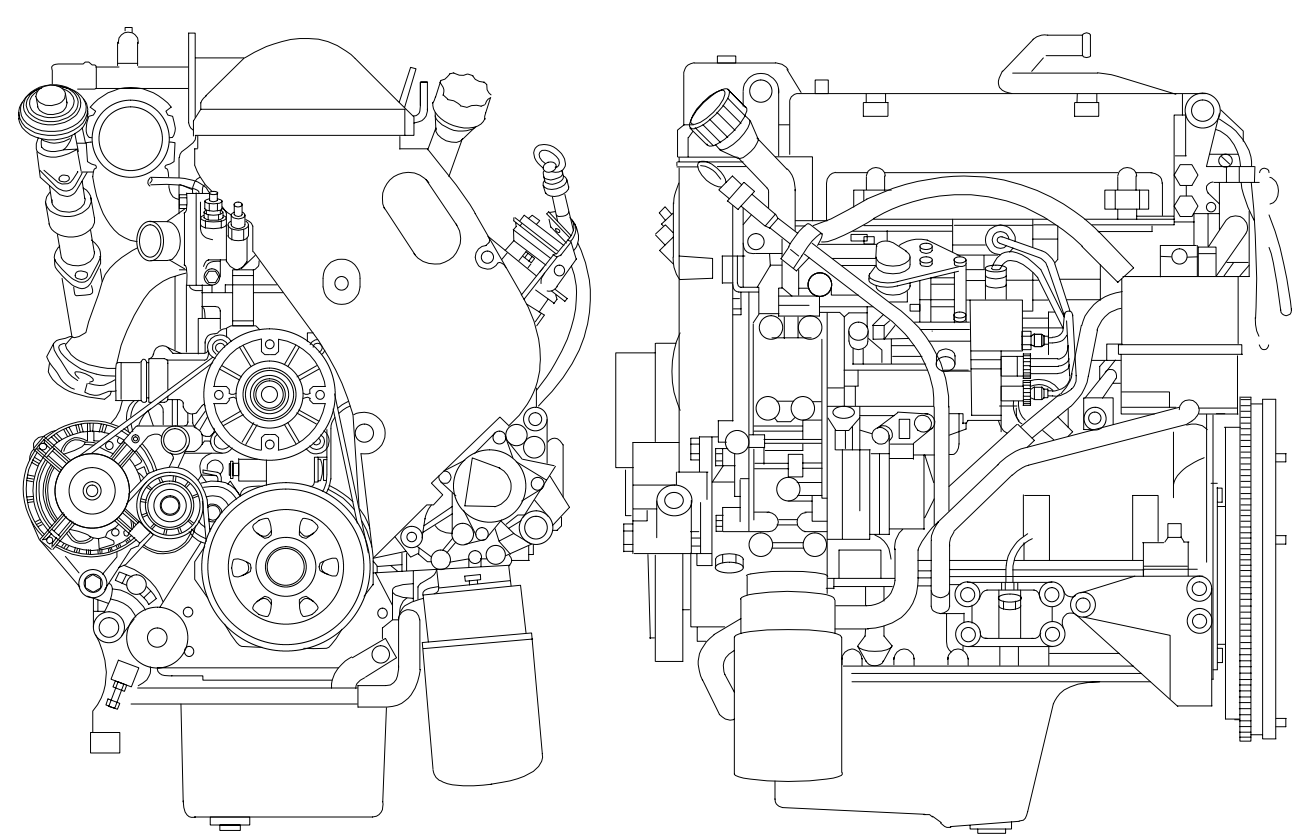
5401 Engine 8140.63.40XX

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EMISSIONS VALUES

Figure 21



52182

Gas emissions

Engine 8140.63 conforms with the ECE 96/69 standard (measurement with vehicle on roller test bench according to ECE + EUDC cycle), with the following limits:

- CO (carbon monoxide) < 1.36 g/km
- HC+NO_x (hydrocarbons + nitrogen oxide) < 1.20 g/km
- Particulate < 0.14 g/km

Test conditions: assigned coast-down 12.7 kW at 80 km/h. Reference weight of vehicle 2270 kg, vehicle equipped with EGR system and oxidation catalyst:

- Degussa I.5 I-300 cpsl-V50-2g Pt
- Kemira I.5 I-300 cpsl-7007-1g Pt

Test fuel: CEC RFT 73 - T - 90 - S ≤ 0.05 %.

Smokiness

Engine 8140.63 conforms with the limits of smokiness required by EEC standards 72/306 and R 24-03 with the following exhaust smokiness values:

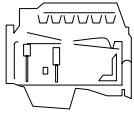
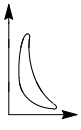
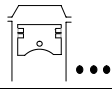
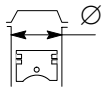
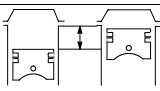
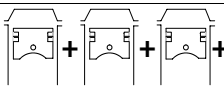
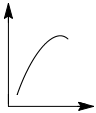
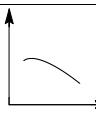
- Maximum power (Bosch BSU opacimeter degrees) 2.0
- Maximum torque (Bosch BSU opacimeter degrees) 3.5
- Full load at 1000 rpm (Bosch BSU opacimeter degrees) 3.5

Noise emissions

Maximum mean noise level, L_{pa}, of the standard engines measured according to ISO Std. 3745 (microphones at 1 m from the engine surfaces):

- Idling (800 rpm) dB A 79
- Full power (3800 rpm) dB A 99

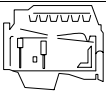
GENERAL SPECIFICATIONS

	Type	8140.63.4.. PC/NA	
	Cycle	Diesel 4 strokes	
	Feeding	Aspirated	
	Injection	Indirect	
	No. of cylinders	4 on-line	
	Diameter	mm	94.4
	Stroke	mm	100
	Total displacement cm ³		2800
	Compression ratio	21.7 ± 0.5	
	Max. power	kW (HP)	58 ÷ 62 (78.9 ÷ 84.3)
		rpm	3800
	Max. torque	Nm (kgm)	179 18.2
		rpm	2000
	Engine idling speed, no load	rpm	800 ± 25
	Maximum engine speed, no load	rpm	4400 ± 25
	Pressure at T.D.C.	*bar	20 ÷ 26
	Minimum permissible pressure at T.D.C.	*bar	16

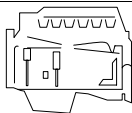
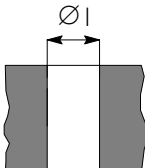
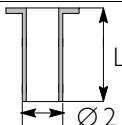
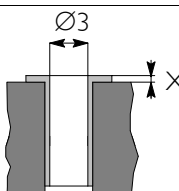
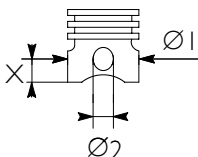
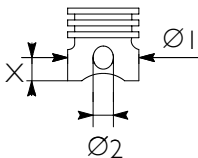
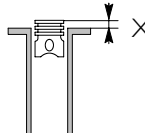
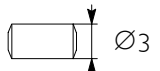
(*) The pressure value is recorded by turning the engine over with the electric starter motor, with oil temperature at 40°- 50°C and the injection pump in the stop condition.

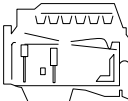
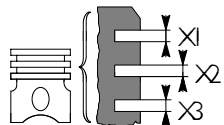
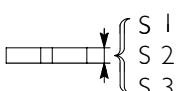
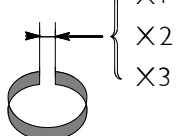
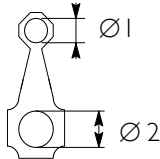
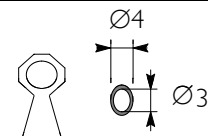
ID = Direct injection
 NA = Aspirated
 PC = Indirect injection (pre-chamber)
 TCA = Supercharging with intercooler

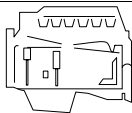
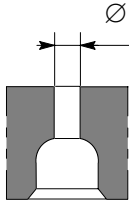
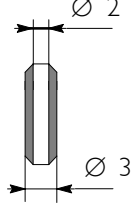
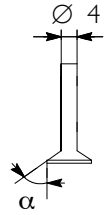
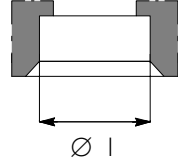
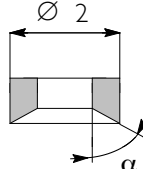
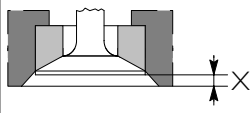
	Type	8140.63.4.. PC/NA
	VALVE TIMING opens before T.D.C. A closes after B.D.C. B opens before B.D.C. D closes after T.D.C. C	 8° 48° 48° 8°
	For timing check Running X mm X mm X mm X mm	 0.5 ± 0.05 0.5 ± 0.05 0.5 ± 0.05 0.5 ± 0.05
	FUEL SUPPLY	
	Bosch type injection pump	rotating distributor VER 808 - 2 (with immobilizer)
	Speed governor Cold start	min. and max. K.S.B. electrical, mechanical
	Pump arrangement With piston n. 1 at T.D.C. Start of delivery mm	 0.94 ± 0.05
	BOSCH injector nozzle type	KGM DNO SACLESS
	Injection order - injection pump - engine	 A - B - C - D 1 - 3 - 4 - 2
	Injection pressure bar	 $130 + 8$
* When checking, for any values of less than 200 bars, calibrate the injectors at $230 + 8$ bar.		

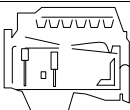
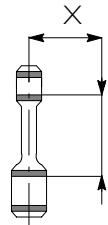
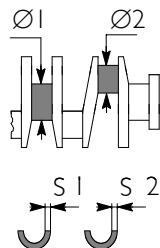
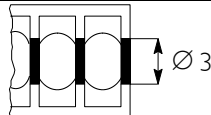
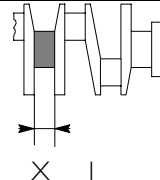
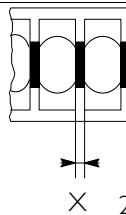
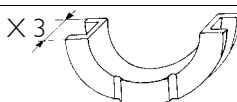
		Type	8140.63.4.. PC/NA
 	LUBRICATION		Forced feed by gear pump, relief valve, dual action oil filter
	Oil pressure, engine hot at idling speed bar at maximum speed bar		≥ 0.8 ≥ 3.5
COOLING		Thermostat: starts to open: fully open:	With a centrifuge pump, driven by the crankshaft via a poly-V belt, thermostat for adjustment, fan with electro-magnetic clutch, radiator. $79^{\circ} \text{ C} \pm 2^{\circ} \text{ C}$ 110° C
 Urania Daily Urania Turbo Urania LD 5	OIL REPLENISHMENT		
	Total capacity at 1st filling		
		liters	6.9
		kg	6.1
	Quantity at periodical replacements:		
	- engine sump		
		liters	5.2
		kg	4.6
	- engine sump		
		liters	6.3
		kg	5.7

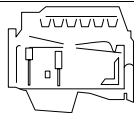
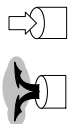
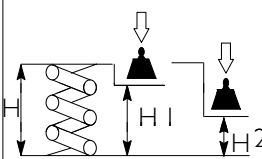
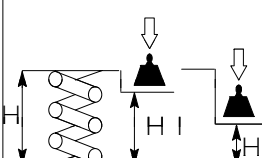
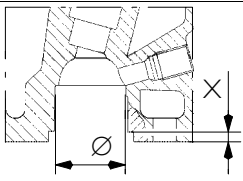
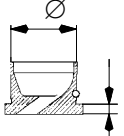
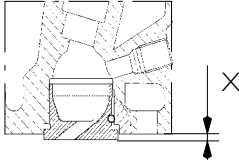
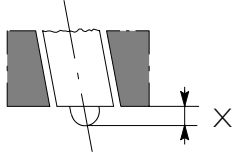
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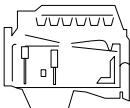
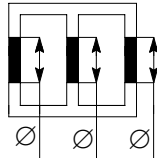
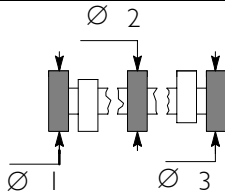
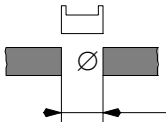
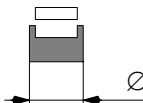
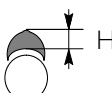
	Type	8140.63.4.. PC/NA	
CYLINDER BLOCK AND CRANK MECHANISM COMPONENTS			
	Bores for cylinder liners Ø 1	97.39 ÷ 97.45*	
	Cylinder liners: outside diameter Ø 2 length L	** 97.47 ÷ 97.50 167.00 ÷ 167.30	
	Cylinder liners - crankcase bores (negative allowance)	0.02 ÷ 0.11	
	Outside diameter Ø 2	0.2	
	Cylinder barrels: (protrusion from engine block bottom)	0.005 max	
	inside diameter  Ø 3	94.402 ÷ 94.432	
	Pistons: supplied as spare parts type: measuring dimension X outside diameter Ø 1 pin bore Ø 2	MONDIAL PISTON 13 94.306 ÷ 94.320 32.000 ÷ 32.005	
	Piston - cylinder sleeve	0.082 ÷ 0.126	
	Pistons: supplied as spare parts type: measuring dimension X outside diameter Ø 1 pin bore Ø 2	AE GOETZE 13 94.306 ÷ 94.320 32.000 ÷ 32.005	
	Piston - cylinder sleeve	0.082 ÷ 0.126	
	Piston diameter Ø 1	0.4	
	Pistons protrusion X	0.75 ÷ 1.15	
	Gudgeon pin Ø 3	31.990 ÷ 31.996	
	Gudgeon pin - pin housing	0.004 ÷ 0.015	
* Diameter to be obtained at servicing when fitting spare cylinder barrels. ** Supplied as spare parts.			

	Type	8140.63.4.. PC/NA	
	Piston type X1* Piston ring grooves X2 X3 * measured on Ø of 91.4 mm	MONDIAL PISTON/AE GOETZE 2.685 ÷ 2.715 2.050 ÷ 2.070 2.540 ÷ 2.560	
	Piston rings S 1* S 2 S 3 * measured on Ø of 91.4 mm	2.568 ÷ 2.597 1.970 ÷ 1.995 2.470 ÷ 2.490	
	Piston rings - grooves 1 2 3	0.088 ÷ 0.147 0.55 ÷ 0.100 0.05 ÷ 0.09	
	Piston rings	0.4	
	Piston ring end gap in cylinder liners X1 X2 X3 X1 X2 X3	0.20 ÷ 0.35 0.30 ÷ 0.50 0.30 ÷ 0.55	
	Small end bush housing Ø 1 Big end bearing housing * Ø 2 * spare connecting rod supplied	35.460 ÷ 35.490 60.341 ÷ 60.348	
	Small end bush diameter outside Ø 4 inside Ø 3 Big end bearing shell supplied as spare parts S	35.570 ÷ 35.595 32.010 ÷ 32.020 1.875 ÷ 1.884	
	Small end bush - housing	0.08 ÷ 0.135	
	Piston pin - bush	0.014 ÷ 0.03	
	Piston rings	0.254 ÷ 0.508	

	Type	8140.63.4.. PC/NA	
CYLINDER HEADS - VALVE GEAR			
	Valve guide housings in the cylinder heads Ø1	12.950 ÷ 12.985	
	Valve guide Ø 2  Ø 3	8.023 ÷ 8.038 13.012 ÷ 13.025	
	Valve guides and housings in the cylinder heads	0.027 ÷ 0.075	
 	> Valve guide	0.05 - 0.10 - 0.2	
	Valves:  Ø 4 α  Ø 4 α	7.985 ÷ 8.000 60° 15' ± 7' 30" 7.985 ÷ 8.000 45° 30' ± 7' 30"	
	Valve stem and its guide	0.023 ÷ 0.053	
	Housing in head for valve seat  Ø1  Ø1	44.025 ÷ 44.075 37.380 ÷ 37.415	
	Outside diameter of valve seat; angle of valve seat in cylinder head:  Ø 2 α  Ø 2 α	44.145 ÷ 44.160 60° ± 5' 37.495 ÷ 37.510 45° ± 5'	
	Recessing of valve X  X 	1.15 ÷ 1.45 1.0 ÷ 1.3	

	Type	8140.63.4.. PC/NA	
CYLINDER BLOCK AND CRANK MECHANISM COMPONENTS			
	Measuring dimension	X	125
	Max. connecting rod axis misalignment tolerance	=	0.07
	Main journals n° 1 - 2 - 3 - 4 n° 5	Ø 1	80.182 ÷ 80.208 86.182 ÷ 86.208
	Crankpins	Ø 2	56.515 ÷ 56.538
	Main bearing shells	S1*	2.165 ÷ 2.174
	Big end bearing shells	S2*	1.875 ÷ 1.884
	* supplied as spare parts		
	Main bearings n° 1 - 2 - 3 - 4 n° 5	Ø 3	84.588 ÷ 84.614 90.588 ÷ 90.614
	Bearing shells - main journals		0.032 ÷ 0.102
	Bearing shells - big ends		0.035 ÷ 0.083
	Main bearing shells		0.254 ÷ 0.508
	Big end bearing shells		0.254 ÷ 0.508
	Main journal, thrust bearing	X 1	31.000 ÷ 31.100
	Main bearing housing, thrust bearing	X 2	26.500 ÷ 26.550
	Thrust washer halves	X 3	30.900 ÷ 30.950
	Crankshaft end float		0.060 ÷ 0.310

	Type	8140.63.4.. PC/NA			
CYLINDER HEADS - VALVE GEAR					
	Between valve seat and head		0.070 ÷ 0.145		
			0.080 ÷ 0.130		
	 > Valve seats	-			
	Valve outside spring height:				
	free height	H	52		
	under a load of:				
	kg 43.8 ± 2.5	H1	38.5		
kg 77.4 ± 4	H2	28.5			
	Valve inside spring height:				
	free height	H	45.5		
	under a load of:				
	kg 16.4 ± 1	H1	33.5		
kg 30 ± 1.5	H2	23.5			
	Height of precombustion chamber plug base seat	X } A B C Ø	5.348 ÷ 5.373		
			5.373 ÷ 5.398		
			5.398 ÷ 5.423		
	Precombustion chamber plug seat		Ø	32.490 ÷ 32.540	
	Precombustion chamber plug	Ø	32.550 ÷ 32.570		
	S } A Green B - C Yellow		5.373 ÷ 5.397		
			5.398 ÷ 5.422		
			5.423 ÷ 5.447		
		Precombustion chamber plug - Cylinder heads	0.010 ÷ 0.080		
	Precombustion chamber position from cylinders heads level surface	-0.000 ÷ +0.049			
	Injector protrusion	X	-		

	Type	8140.63.4.. PC/NA	
CYLINDER HEADS - VALVE GEAR			
	Camshaft bearing housing		
	normal	Ø	33.985 ÷ 34.015
	oversized	Ø	34.185 ÷ 34.215
	Camshaft bearing journals		
	normal	Ø	33.934 ÷ 33.950
	oversized	Ø	34.134 ÷ 34.150
	Between seats and supporting pins		0.035 ÷ 0.081
	Tappets housing on cylinder heads		
	Ø normal		44.000 ÷ 44.025
	Tappet	Ø normal	43.950 ÷ 43.970
	Between tappets and seats		0.030 ÷ 0.075
	Cap		3.25 to 4.45 mm with a progression of 0.05
	Cam lift:		
	 H	H	10.5
	 H	H	10.5

TIGHTENING TORQUES

NOTE This differs from the analogous section for the 8140.43C engine in the following tightening torques.

PART	TORQUE	
	Nm	kgm
Nut M12x1.5 fixing toothed bushing to the injection pump	55	5.5
Fixing incandescent glow plugs	25	2.5
Nut fixing electric cable to the glow plug	1.5 ÷ 2	0.15 ÷ 0.2
Injector fastening	60	6

TOOLS

NOTE This differs from the analogous section for the 8140.43C engine in the exclusion of tools 99360486 - 99367121 - 99394038 - 99395098 and the addition of the following tools.

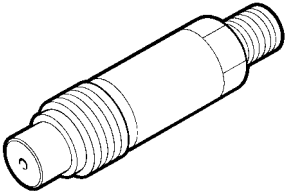
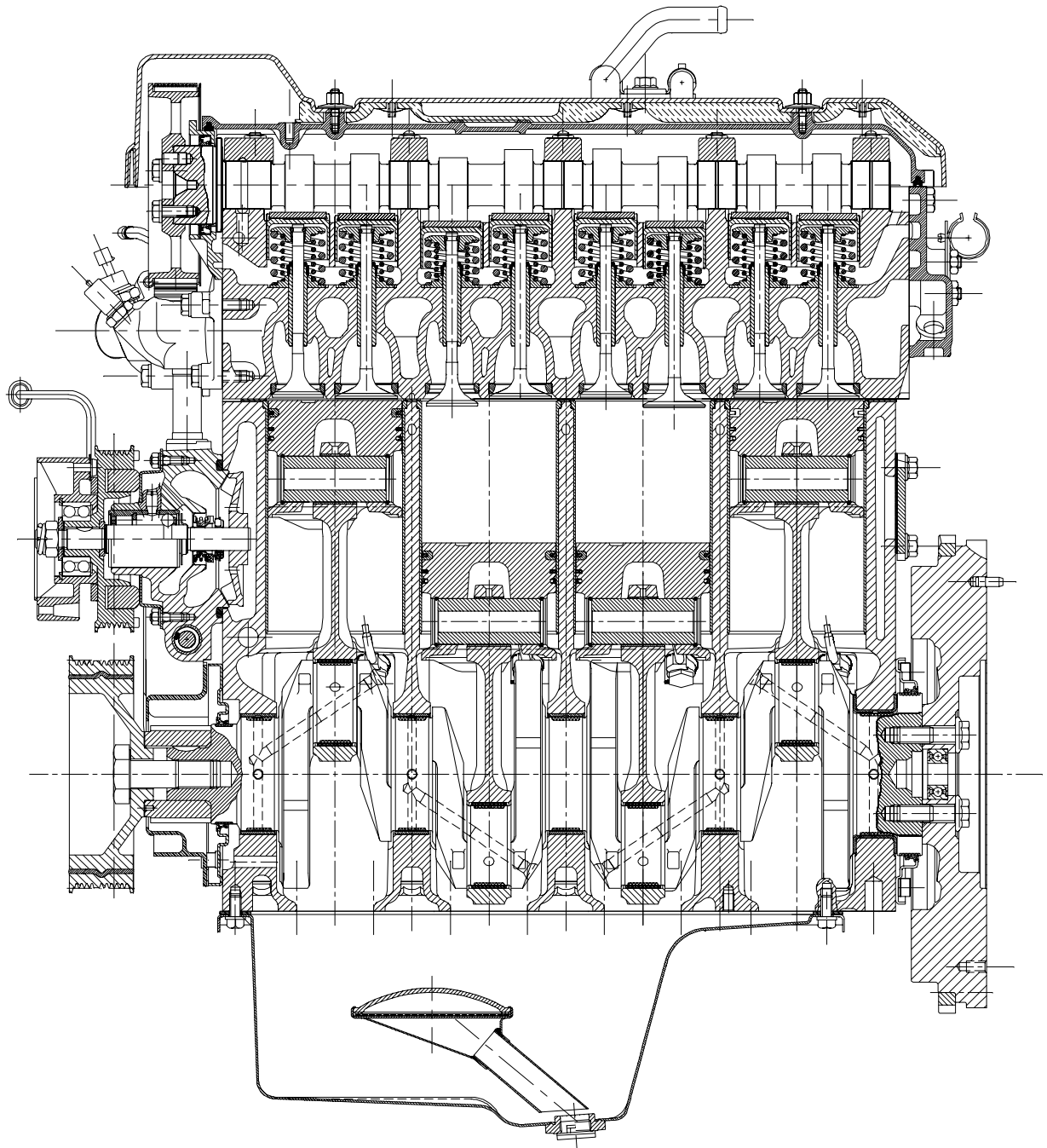
TOOL NO.	DESCRIPTION
99360498	 Cylinder compression test coupling (use with 99395682)

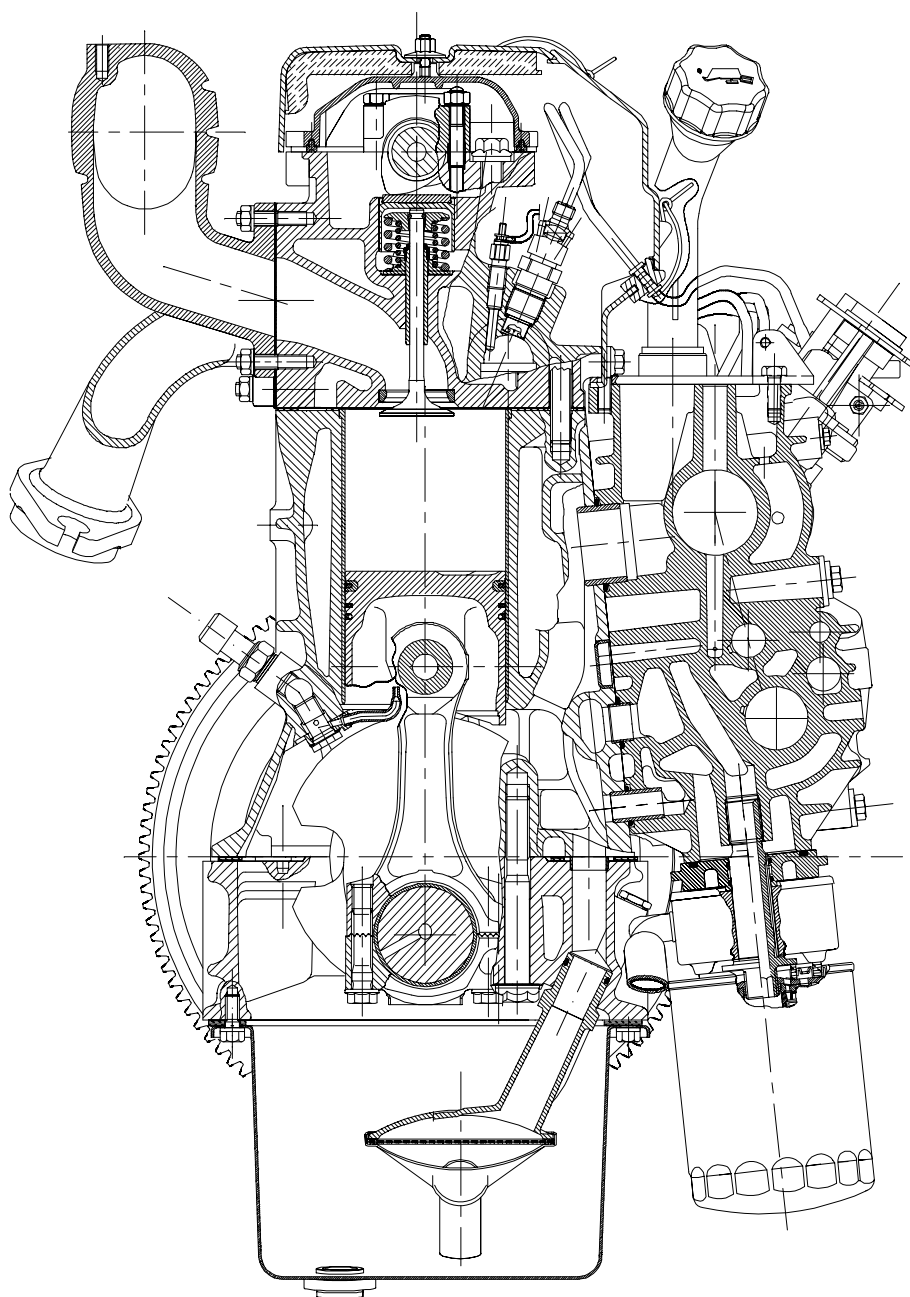
Figure 22



52180

8140.63.4... ENGINE LONGITUDINAL SECTION

Figure 23



8140.63.4... ENGINE TRANSVERSE SECTION

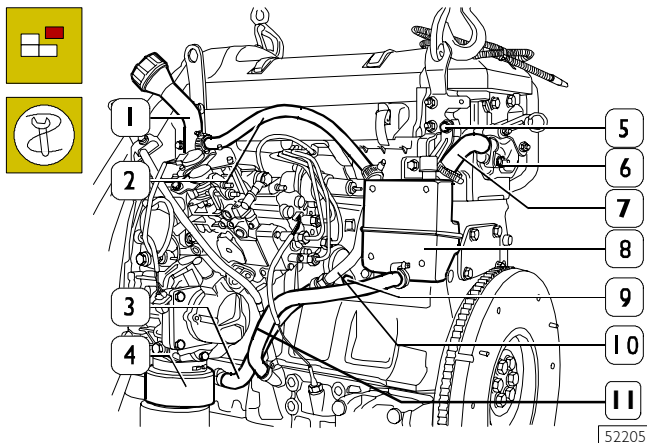
52181

8140.63 ENGINE OVERHAUL

NOTE This differs from overhauling engine 8140.43.C in the following.

Disassembly engine on bench

Figure 24

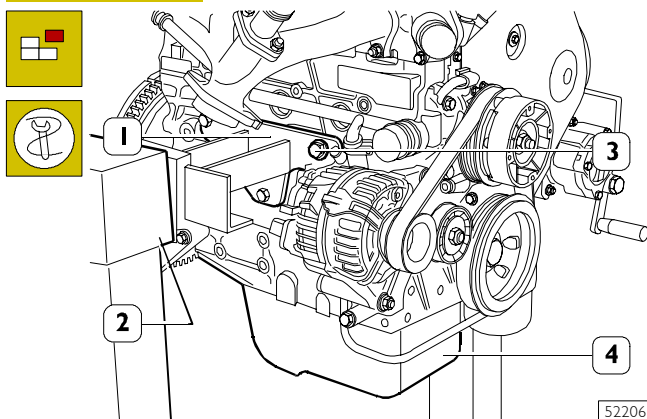


To be able to apply the brackets 99361029 to the crankcase for securing the engine to the stand for overhauling, it is necessary:

on the left-hand side of the engine:

- ☐ to disconnect the pipe (3) from the heat exchanger (4);
- ☐ to disconnect the pipe (7) from the side cover (6);
- ☐ to take out the screws (5 and 9) and remove the pipe (10) complete with pipes (3 and 7);
- ☐ to disconnect the pipe (11) from the crankcase fitting;
- ☐ to disconnect the pipe (2) from the fillpipe (1);
- ☐ to remove the oil vapour condenser (8) complete with pipes (2 - 10 - 11).

Figure 25

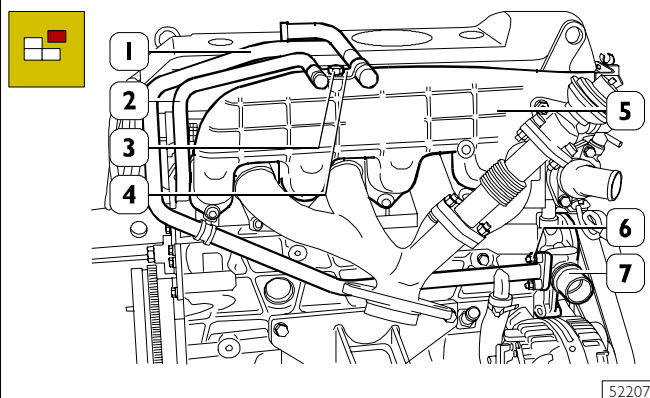


On the right-hand side, take the plug out of the crankcase to be able to screw down the screw (3) fixing the bracket 99361029 (1) and to mount this bracket.

Mount the other bracket on the left-hand side of the crankcase and secure them to the rotary stand 99322230 (2).

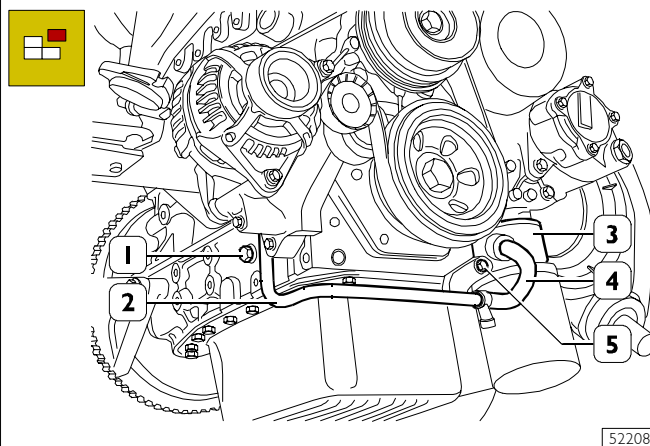
Take the plug out of the oil sump (4) and drain off the lubricating oil.

Figure 26



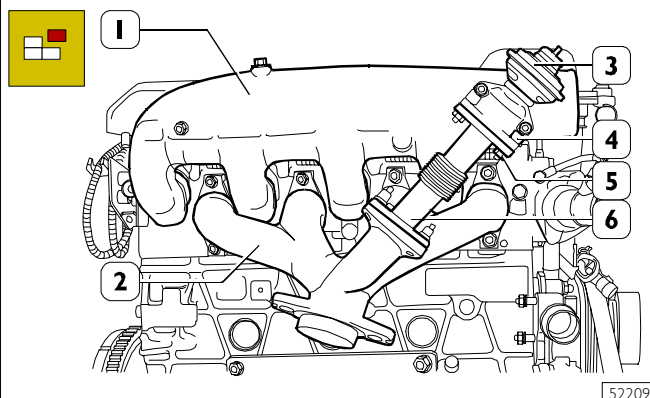
Take out the screw (3) fixing the bracket (4) securing the pipes (1 and 2) to the intake manifold (5). Disconnect the pipe (2) from the cylinder head. Take out the nuts (6) securing the pipe (1) to the water pump (7).

Figure 27



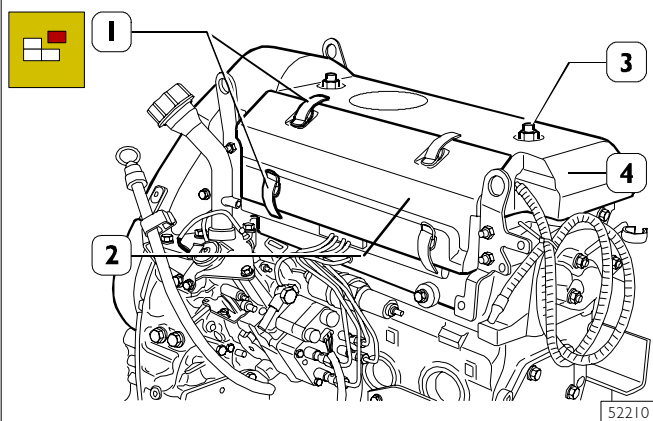
Disconnect the pipe (4) from the heat exchanger (3). Take out the screws (1 and 5) and remove the pipe (2) from the crankcase complete with pipes (4) and pipes (1, Figure 26).

Figure 28



Take out the nuts (5) for the screws (4) and disconnect the EGR valve (3) from the sleeve (6). Remove the intake manifold (1) and exhaust manifold (2) from the cylinder head.

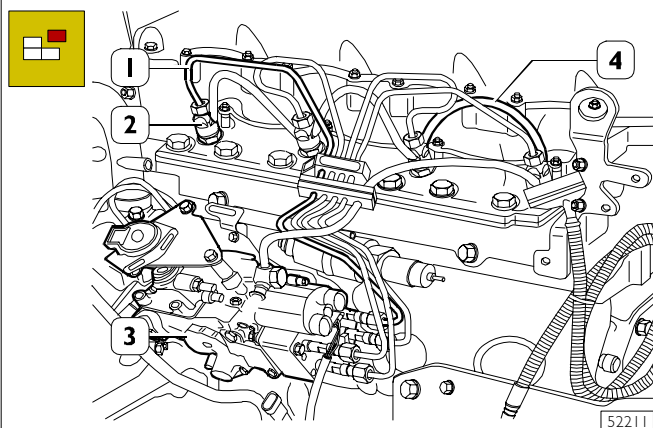
Figure 29



Lift the clips (1) and remove the side soundproofing guard (2).

Take out the nuts (3) and remove the top soundproofing guard (4).

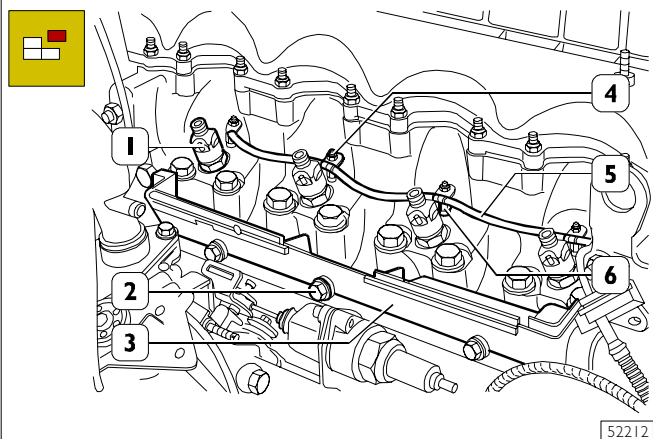
Figure 30



Remove the fuel pipes (1) from the injection pump (3) and from the injectors (2).

Remove the fuel recovery pipes (4) from the injectors (2).

Figure 31

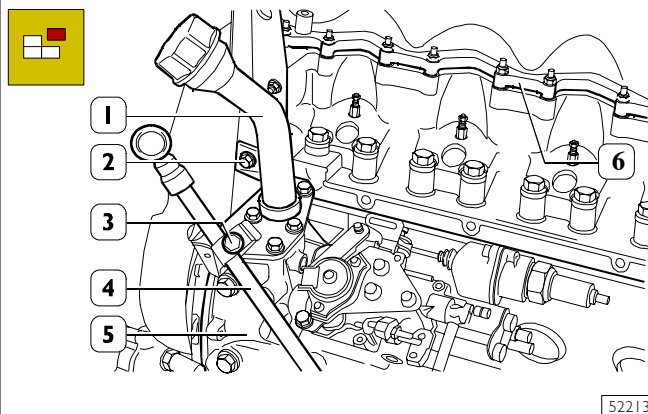


Take out the nuts (4) and disconnect the electric cable (5) from the glow plugs (6).

Unscrew and remove the injectors (1).

Take out the screws (2) and remove the guard (3).

Figure 32

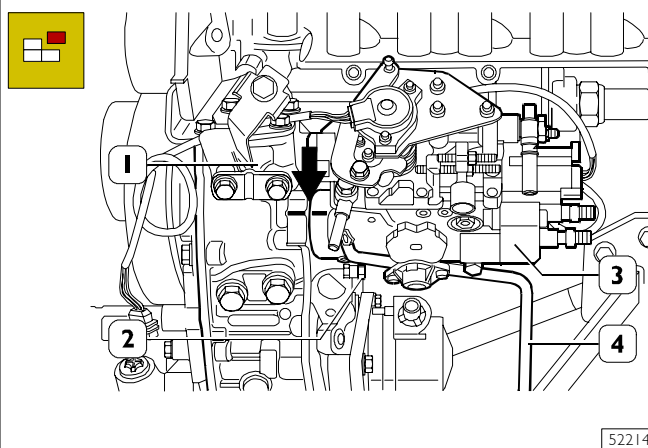


Remove the tappet cover (6) with its relative gasket.

Take out the screw (2) and remove the fillpipe (1) from the auxiliary member assembly (5).

Take out the screw (3) and extract the oil dipstick pipe (4) from the crankcase base.

Figure 33



Mark (⇒) the position of assembly of the injection pump (3) on the auxiliary member assembly (1).

Take out the fixing nuts (2) and remove the injection pump (3) from the auxiliary member assembly (1).

NOTE To remove the lower fixing nut on the crankcase side, use wrench 99352114 (4).